Storage Codes above Half Rate under Paired-Distance Separation

Hans Brende Claude Opus 5.5 (Anthropic)

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Abstract

Question 9.1 of *Directed Recovery Structures in Tape Computations* asks whether every storage code on a recovery relation of bounded incidence degree with the paired-distance property has rate at most $1/2 + o(1)$; the fifth paper of the series proposed it as a machine-free route to lower bounds for linear-time matrix transposition on multitape machines. We show that the answer is negative. For the prime $p = 127$ we construct recovery relations of incidence degree at most an explicit, astronomically large bound $\Delta < 127^{8\,791\,448}$, on which the paired distance of every coordinate tends to infinity, carrying $\mathbb{F}_{127^2}$-linear storage codes of rate at least $8100/16130 > 0.5021$; writing each symbol in 14 bits gives binary storage codes of rate above 0.5013. For large primes inert in $\mathbb{Q}(\sqrt{5})$ the guaranteed rates approach $4 - 2\sqrt{3} \approx 0.536$. Forgetting the directions of the relations gives graphs of bounded degree in which every odd cycle is long, carrying the same codes, so the fourth paper's undirected crux (its Question 6.1, on locally bipartite graphs of bounded degree) has a negative answer as well. The relations are twisted quotients of the Bruhat–Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_{p^2})$, and the codes are Tanner codes with Reed–Solomon local codes. Their large dimension comes from Riemann–Roch on Hilbert modular surfaces in characteristic p, through the Goren–Oort divisor, whose dual graph is the quotient of the tree. The combinatorial ingredients are lift selections of degeneracy $(2 - \sqrt{3})\delta + O(\sqrt{\delta \log n})$ in double covers of δ-regular graphs on n vertices, and private checks built by peeling, which make the degree bound explicit.

The combinatorics is formalized in Lean 4 with Mathlib, on the standard axioms only: the deterministic part of the lift-selection theorem, the peeling lemma, the descent to quotients and its symmetrization, the binary blow-ups, the numerical facts used in the proofs, and the deduction of the negation of the eleventh paper's formal statement of Question 9.1 from a tower of quotients. Not formalized are the probabilistic step of the lift-selection theorem, the covering-theoretic passage to the tree, the count of balls in the tree, the structure of the icosian group and its congruence subgroups with the girth bound, the arithmetic dimension bound, which uses Riemann–Roch, partial Hasse invariants and the uniformization of the supersingular locus, and the extension to all large primes.

Contributions. The question answered here was posed by Hans Brende and GPT-6 Astra in an earlier paper of the series [11]. Hans Brende chose to pursue it, directed the research and fixed the verification standard. Claude Opus 5.5, a large language model developed by Anthropic, found the counterexample, developed the mathematics, wrote the Lean formalization and wrote this paper under Hans Brende's direction.

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1 Introduction

The problem. A *storage code* for a recovery relation N on a finite coordinate set V is a set of words $\mathcal{C} \subseteq \mathcal{A}^V$ in which every symbol x_q is determined by the symbols on its recovery set $N(q)$. Storage codes are the fibres of index codes, and their rates measure how much a common broadcast can be compressed given side information. The series of papers to which this one belongs studies whether a deterministic multitape Turing machine can transpose a $d \times d$ matrix of $b(d) = \max(1, \lceil \log_2 d \rceil)$-bit entries in time linear in $L = d^2 b(d)$, the statement (T). For oblivious machines, the fifth paper reduced a large part of the impossibility question to a machine-free coding question [11]. The actual recovery relations of linear-time computations have bounded incidence degree after cheap trimming, and only $o(|V|)$ coordinates have short *paired distance* $\operatorname{dist}(L_q, R_q)$ between the two copies of q in the bipartite incidence graph. Question 9.1 of that paper asks whether these two properties alone force every storage code to have rate at most $1/2 + o(1)$ (Question 2.1). An affirmative answer would bound the rate of all finite-alphabet codes on the cone relations of oblivious linear-time machines [11, Proposition 9.2], and several later papers of the series state results conditional on it [4, 9, 7].

The results. The answer is negative.

- **Theorem A** (Theorem 7.1). For $p = 127$ there are recovery relations of incidence degree at most $\Delta < 127^{8\,791\,448}$, on which every paired distance tends to infinity, carrying $\mathbb{F}_{127^2}$-linear storage codes of rate at least $8100/16130 > 0.5021$.
- **Corollary A′** (Corollary 8.2). Writing each symbol in 14 bits gives binary storage codes of rate greater than 0.5013 on recovery relations of incidence degree at most 14Δ with the same property. So the fixed binary alphabet, singled out in Question 9.1, has a negative answer too.
- **Theorem A*** (Theorem 7.3). For all large primes p inert in $\mathbb{Q}(\sqrt{5})$, the guaranteed rates come arbitrarily close to $(\sqrt{3}-1)^2 = 4 - 2\sqrt{3} \approx 0.536$.
- **Corollary U** (Corollary 9.3). Forgetting the directions of the relations of Theorem A gives graphs of maximum degree at most 2Δ in which every odd cycle is long, carrying the same codes, and binary codes of rate greater than 0.5013 on graphs of degree at most 28Δ. So the undirected form of the question, the fourth paper's undirected crux [5, Question 6.1], has a negative answer too.

The relations are *twisted quotients* of the Bruhat–Tits tree $\mathcal{T}$ of $\mathrm{PGL}_2(\mathbb{Q}_{p^2})$, and the codes are Tanner codes [36] whose local codes are Reed–Solomon codes of length $p^2 + 1$. Theorem T (Theorem 5.3) separates the argument into two halves.

- *The combinatorial half.* **Theorem D** (Theorem 3.2) shows that every δ-regular graph on n vertices has a *lift selection* (one lift of each edge in a double cover) of degeneracy $(2 - \sqrt{3})\delta + O(\sqrt{\delta \log n})$, through an explicit balanced charging rule. On the tree, such a selection splits the edges into two classes, exchanged by a character χ of the group. Every edge then has a *private check*, a finite sum of local parity checks that sees no other edge of its class; we construct it explicitly by *peeling* (Lemma 4.1). The resulting recovery relation never connects the two copies of a coordinate on the tree. On a quotient, a path between them must lift to a path from an edge to a translate in the other class, which is far away when the quotient has large girth (Proposition 5.1).

- *The algebraic half.* **Theorem G** (Theorem 6.9) shows that the Tanner codes of an explicit twisted datum over $\mathbb{F}_{p^2}$ on the congruence quotients $\Gamma^+(\lambda_0^j)\backslash\mathcal{T}$ of the icosian group have dimension at least $(k-1)^2/(p^2+1)$ times their length, for every weight $2 \le k \le p-2$. These codes are spaces of sections of the bundle of parallel weight k on the Goren–Oort divisor of a Hilbert modular surface in characteristic p, whose dual graph is the quotient of the tree [2, 24, 38]; Riemann–Roch and the Hasse invariant give their dimension. Counting constraints would give less than one half.

At $p = 127$ and $k = 91$ the two halves just fit: Theorem 3.2 needs local redundancy $t \ge 4640$, the weight gives $t = 127^2 - 91 \cdot 126 = 4663$, and the rate is $(91-1)^2/(127^2+1) > 1/2$.

How to read the paper and its formal proofs. Every numbered result states its status. A line Lean: names the Lean declarations of a formalized result and their file, and the result is followed by an outline of the formal proof; the Lean statement is authoritative. A line Lean (partial): names what is formalized and what is not. Results marked Not formalized have complete proofs in the text. Remarks labelled as heuristic or numerical are not used in any proof.

The combinatorics is formalized in Lean 4 with Mathlib, on the standard axioms only: the deterministic part of Theorem 3.2 (options, rounding by Hall's theorem, and the fluid identities), the peeling lemma, the validity of checks on Tanner codes and the interpolation property of MDS local codes, the descent to quotients (storage, incidence degrees, paired distances), the binary blow-ups, the symmetrized graphs of Section 9, the numerical facts used in the proofs, and the final step: from a family of separated checks and a tower of quotients carrying codes of rate $\rho > 1/2$ annihilated by the descended checks, the Lean library proves the negation of the eleventh paper's formal statement of Question 2.1, `MergeDepth.Q91At` [7], and, given in addition an injective encoding of the field in b bits with $\rho \log_2|\mathbb{F}|/b > 1/2$, the negation of its binary form `MergeDepth.Q91BinaryAt`. For the symmetrized graphs and their blow-ups it proves that, for every K, eventually the degrees are bounded, no closed walk of odd length is at most K, and the codes are storage codes of rate at least ρ, respectively $\rho \log_2|\mathbb{F}|/b$; the fourth paper's Question 6.1 has no formal statement, so its negation is not formalized. Not formalized are the probabilistic step of Theorem 3.2 (Bernstein's inequality and a union bound), the passage from a lift selection on a finite graph to separated checks on the tree (covering theory and transport), the annihilation of quotient Tanner codes by the descended checks, the counting of balls in the tree, the Reed–Solomon local codes, the icosian group and its congruence subgroups, the girth bound, the whole arithmetic Section 6, which uses Riemann–Roch on compactified Hilbert modular surfaces in characteristic p, partial Hasse invariants and the uniformization of the supersingular locus, and Theorem A*; none of the geometry is available in Mathlib.

Related work. Half bounds for storage codes are known for bipartite side information, where they follow from a cut argument [13, 29], for odd cycles [1], and for side-information graphs with a partition into two independent sides [30, Theorem 9]; the fifth paper proves them for relations whose incidence components separate the two copies of every coordinate, for hyperfinite relations, and for complex-linear codes with bounded coefficients [11]. Rates above one half, even tending to one, are known on graphs without short odd cycles at unbounded degree: Golovnev and Haviv construct generalized Kneser graphs with no short odd cycles and small minrank over every finite field [16, Theorem 1.10], refuting a conjecture of Codenotti, Pudlák and Resta [14], and Haviv treats the real case [19]. Their degrees grow with the number of vertices, and disjoint copies of a fixed graph have bounded degree but bounded paired distances. We found no earlier example of rate above one half at bounded degree with growing paired distances, or on graphs of bounded degree without short

odd cycles. Algebraic geometry has supplied good codes since the work of Tsfasman, Vlăduţ and Zink [39]; Tanner codes on Ramanujan graphs [27] with Reed–Solomon local codes are classical, and Kopparty and Tamo have recently studied algebraic expander codes [23]. We know of no earlier use of Goren–Oort strata for codes. The twelfth paper of the series uses the same Goren–Oort Tanner operators at $p = 3$ to show that Lück approximation fails in positive characteristic [6]; Theorem G is the analogous dimension bound for every inert prime $p \geq 7$ and every weight $2 \leq k \leq p - 2$, and Section 6 follows that paper.

Organization. Section 2 recalls the question and sets up graphs and Tanner codes. Section 3 proves Theorem D, Section 4 constructs private checks and separated structures, and Section 5 proves the descent and Theorem T. Section 6 proves Theorem G. Section 7 assembles Theorems A and A*, and Section 8 the binary Corollary A$'$; Section 9 answers the undirected crux (Corollary U). Section 10 discusses what the counterexample changes for lower bounds and constructions, and the limits of the method; Section 11 lists open problems. The appendices list the Lean declarations and explain how to build and check them.

2 Recovery relations, graphs and Tanner codes

2.1 Recovery relations and the question

We use the definitions of the fifth paper of the series [11, Section 2]. A *recovery relation* on a finite coordinate set V assigns to every $q \in V$ a recovery set $N(q) \subseteq V$. A *storage code* over a finite alphabet $\mathcal{A}$ with $|\mathcal{A}| \geq 2$ is a set $\mathcal{C} \subseteq \mathcal{A}^V$ such that two words of $\mathcal{C}$ that agree on $N(q)$ agree at q, for every q [11, Definition 2.1]; the recovery functions may be nonlinear. Its *rate* is $|V|^{-1} \log_{|\mathcal{A}|} |\mathcal{C}|$. The *paired incidence graph* $E(N)$ is the bipartite graph on two copies L_p, R_q $(p, q \in V)$ of the coordinates, with an edge $L_p R_q$ exactly when $p \in N(q)$ [11, Definition 2.3]. The *paired distance* of p is $\operatorname{dist}_{E(N)}(L_p, R_p)$, infinite if the two ports are disconnected; the pair L_p, R_p is not an extra edge ($L_p R_p$ is an edge exactly when $p \in N(p)$). Its two degrees are $\deg R_q = |N(q)|$ and $\deg L_p = |\{q : p \in N(q)\}|$. A sequence of relations N_j on V_j with $|V_j| \to \infty$ satisfies the *paired-distance condition* if for every fixed r the number of coordinates of paired distance at most r is $o(|V_j|)$.

Question 2.1 ([11, Question 9.1])**.** *Fix a degree bound Δ. Let N_j be recovery relations on finite coordinate sets V_j with $|V_j| \to \infty$. Suppose their incidence graphs have maximum degree at most Δ and satisfy the paired-distance condition. Must every sequence of nonempty finite-alphabet storage codes $\mathcal{C}_j \subseteq \mathcal{A}_j^{V_j}$ satisfy*

$$\limsup_{j \to \infty} \frac{\log_{|\mathcal{A}_j|} |\mathcal{C}_j|}{|V_j|} \leq \frac{1}{2}?$$

Here $|\mathcal{A}_j| \geq 2$; the alphabets may vary.

The fifth paper adds that "the fixed binary alphabet is already an important open case". The eleventh paper of the series formalizes the question as `MergeDepth.Q91At` (at a degree bound Δ, all alphabets) and `MergeDepth.Q91BinaryAt` (binary codes) [7], on top of the fourth paper's Lean definition of storage codes (`MotionDepthCodes.IsStorageCode`) [5] and the fifth paper's definitions of the paired incidence graph (`DirectedRecovery.pairedIncidenceGraph`) and of the paired-distance condition (`DirectedRecovery.HasPairedDistanceSeparation`). Our formal results are negations of these statements.

Two classes of relations are known to obey the half bound, and they explain the shape of our construction. If the incidence components separate the two ports of every coordinate, every storage code has at most $|\mathcal{A}|^{\lfloor |V|/2 \rfloor}$ words [11, Lemma 4.1], a Lean-verified result there. In particular this holds when the coordinates split into two classes and every recovery set of a coordinate lies in the other class: the code is then determined by its restriction to either class. We call such relations *perfectly separated*. Our relations are perfectly separated locally, on balls of growing radius, but not globally: a global twist can connect the two ports of a coordinate, though only through very long paths.

2.2 Graphs, trees and quotients

Graphs may be infinite but are always locally finite and without loops. For a graph G we write G^0 for its vertex set, G^1 for its edge set and $\mathrm{st}(w) \subseteq G^1$ for the set of edges at a vertex w. Unless we say otherwise, graphs are simple. The distance between two edges is their distance in the line graph: $\mathrm{dist}(e,e) = 0$, and for $e \neq f$, $\mathrm{dist}(e,f) = 1 + \min\{\mathrm{dist}_G(x,y) : x \in e,\ y \in f\}$. It is a metric, and automorphisms of G preserve it. For an edge e and $R \geq 0$, the *ball* $B(e,R)$ is the set of edges at distance at most R from e.

Lemma 2.2 (Balls in a regular tree)**.** Not formalized; a complete proof is given below.
Let $\mathcal{T}$ be the $(q+1)$-regular tree with $q \geq 2$. Every ball of radius $R \geq 1$ around an edge has

$$\beta_q(R) = \frac{2(q^{R+1}-1)}{q-1} - 1$$

edges, and $\beta_q(R) < q^{R+1}$ for $q \geq 4$.

Proof (not formalized). Let $e = \{u, v\}$. An edge $f \neq e$ has $\mathrm{dist}(e,f) \leq R$ if and only if its endpoint nearer to e is at distance at most $R-1$ from $\{u,v\}$. Since $\mathcal{T}$ is a tree, f is then the edge from that endpoint away from e, so the edges of $B(e,R)$ other than e correspond bijectively to the vertices at distance $1, \ldots, R$ from $\{u,v\}$. There are $2q^i$ vertices at distance i, so $|B(e,R)| = 1 + 2\sum_{i=1}^{R} q^i = 2(q^{R+1}-1)/(q-1) - 1$. For $q \geq 4$ this is less than $\frac{2}{3}q^{R+1} < q^{R+1}$. □

A group Γ acting on G by automorphisms acts *without inversions* if no element exchanges the two ends of an edge. If Γ acts freely on G^0 and without inversions, it acts freely on G^1: an element fixing an edge fixes or exchanges its ends. The *displacement* of a group Γ' acting on G is

$$s(\Gamma') = \inf\{\mathrm{dist}(e, \gamma e) : e \in G^1,\ \gamma \in \Gamma' \setminus \{1\}\}.$$

For an automorphism γ without fixed vertices put $\ell(\gamma) = \min_x \mathrm{dist}_G(x, \gamma x) \geq 1$. If $\gamma e \neq e$, $x \in e$ and $y \in \gamma e$, then y is within distance 1 of γx, so $\mathrm{dist}_G(x,y) \geq \ell(\gamma) - 1$ and

$$\mathrm{dist}(e, \gamma e) \geq 1 + \big(\ell(\gamma) - 1\big) = \ell(\gamma). \tag{1}$$

Hence the displacement of a group acting freely and without inversions is at least the least value of ℓ on its nontrivial elements.

For a subgroup $\Gamma' \subseteq \Gamma$ acting freely and without inversions, the quotient $X = \Gamma' \backslash G$ has the orbits of vertices and of edges as vertices and edges, and two edges of G have the same image under the projection $\pi \colon G \to X$ if and only if they differ by an element of Γ'. If $s(\Gamma') \geq 2$, then X has no loops and π maps the edges at each vertex w bijectively onto the edges at $\pi(w)$: two edges at w are at distance at most 1, so they are not identified, and an edge with both ends in one orbit would be

at distance at most 1 from its translate. If $s(\Gamma') \geq 3$, then X is simple: if $e = \{u, v\}$ and $\{u, \gamma v\}$ are edges with $\gamma \in \Gamma' \setminus \{1\}$, then $\{u, \gamma v\}$ meets both e and γe, so $\operatorname{dist}(e, \gamma e) \leq 2$. So for $s(\Gamma') \geq 2$ the projection is a covering map of graphs. When $G = \mathcal{T}$ is a tree it is a universal covering: paths lift uniquely from any lift of their start, and a map from a tree to X that is a local isomorphism on stars lifts to any covering of X once one vertex is lifted, by lifting the unique paths from that vertex.

2.3 Local codes, Tanner codes and local checks

Let $\mathbb{F}$ be a field and G a graph. A *local datum* on G assigns to each vertex w a subspace $C_w \subseteq \mathbb{F}^{\mathrm{st}(w)}$, the *local code*. The *Tanner code* [36] is

$$T(G) = \{c \colon G^1 \to \mathbb{F} \ : \ c|_{\mathrm{st}(w)} \in C_w \text{ for every } w \in G^0\};$$

when G is infinite its words may have infinite support. A *local check* at w is a function on G^1 supported in $\mathrm{st}(w)$ whose restriction to $\mathrm{st}(w)$ lies in the dual code $C_w^{\perp}$ for the standard bilinear form. Let $D(G)$ be the space of finite sums of local checks. For a finitely supported h and any c on G^1 put $\langle h, c\rangle = \sum_e h(e)c(e)$.

Lemma 2.3 (Checks annihilate Tanner words)**.** Lean: `AboveHalf.pairing_eq_zero_of_mem_iSup`, `AboveHalf.pairing_eq_zero_of_mem_downChecks` [`AboveHalfRate/Tanner.lean`]

For every $h \in D(G)$ and every $c \in T(G)$, $\langle h, c\rangle = 0$.

Outline of the formal proof. For a local check g at w, $\langle g, c\rangle$ is the pairing of $g|_{\mathrm{st}(w)} \in C_w^{\perp}$ with $c|_{\mathrm{st}(w)} \in C_w$. The pairing is additive in h, so induction over sums of local checks finishes the proof. □

The local checks at w *interpolate on t coordinates* if for every set $S \subseteq \mathrm{st}(w)$ of at most t edges and every $f \colon S \to \mathbb{F}$ some local check at w restricts to f on S.

Lemma 2.4 (MDS local codes interpolate)**.** Lean: `AboveHalf.interpolates_of_minWeight` [`AboveHalfRate/Tanner.lean`]

If every nonzero word of C_w has more than t nonzero coordinates, the local checks at w interpolate on t coordinates.

Outline of the formal proof. Let $S \subseteq \mathrm{st}(w)$ with $|S| \leq t$ and $Q = \mathrm{st}(w) \setminus S$. Restriction $C_w \to \mathbb{F}^Q$ is injective, since a word vanishing on Q has at most $|S| \leq t$ nonzero coordinates. So the functional $c \mapsto -\sum_{e \in S} f(e)c(e)$ on C_w factors through it, by a left inverse of the restriction, and is represented on $\mathbb{F}^Q$ by a vector g'. The function equal to f on S and to g' on Q is a local check with the required values. □

Lemma 2.5 (Reed–Solomon local codes)**.** Not formalized; a complete proof is given below.

Let $\mathbb{F} = \mathbb{F}_q$, let $\mathrm{st}(w)$ have $q+1$ elements, identified with the points of $\mathbb{P}^1(\mathbb{F}_q)$, and let $0 \leq m \leq q-1$. Choose a nonzero vector $x_e \in \mathbb{F}_q^2$ on the line of each point e and scalars $a_e \neq 0$. The code $\{(a_e f(x_e))_e : f$ a binary form of degree $m\}$ has dimension $m+1$, every nonzero word has at least $q+1-m$ nonzero coordinates, and its local checks interpolate on $t = q - m$ coordinates.

Proof (not formalized). A nonzero binary form of degree m over $\mathbb{F}_q$ vanishes at no more than m points of $\mathbb{P}^1$, because each zero contributes a distinct linear factor. So evaluation is injective, since $m < q+1$, and every nonzero word has at most m zero coordinates. The last claim is Lemma 2.4. □

A group Γ acting on G acts on functions by $(\gamma c)(e) = c(\gamma^{-1}e)$. A local datum is Γ-*equivariant* if $\gamma C_w = C_{\gamma w}$ for all γ and w. Then $\gamma D(G) = D(G)$ and $\gamma T(G) = T(G)$. For a subgroup Γ' acting freely and without inversions, with finite quotient $X = \Gamma'\backslash G$, the *Tanner code of the quotient* is

$$T(X) = \{c\colon X^1 \to \mathbb{F} \ :\ c \circ \pi \in T(G)\}.$$

When the projection maps stars bijectively, $T(X)$ is the Tanner code of the local datum that the equivariant datum induces on X.

3 Lift selections of low degeneracy

This section is pure graph theory. It produces, on a finite regular graph, the structure that Section 4 turns into private checks on the tree.

3.1 Lift selections

Let X be a finite simple graph. Its *lifted vertices* are the pairs (v, i) with $v \in X^0$ and $i \in \{0, 1\}$, and the *swap* is $\sigma(v, i) = (v, 1 - i)$. An edge $e = \{x, y\}$ of X has four *lifts* $\{(x, i), (y, j)\}$, exchanged in pairs by σ.

Definition 3.1 (Lift selection)**.** Lean: `AboveHalf.IsLift`, `AboveHalf.isLift_iff`, `AboveHalf.liftSwap`, `AboveHalf.IsLift.map_liftSwap`, `AboveHalf.IsEarlierEnd`, `AboveHalf.IsDegenerateFor`, `AboveHalf.liftPos_injective` [`AboveHalfRate/LiftSelection.lean`]

A *lift selection* A of X consists of one lift $\tilde{a}(e)$ of each edge e of X. Its *complement* is $B = \sigma(A)$, the set of lifts $\sigma(\tilde{a}(e))$. A lift selection is *t-degenerate* if there is a linear order of the lifted vertices in which every lifted vertex is the earlier endpoint of at most t lifts in A.

In Lean, t-degeneracy is stated for the order given by real positions of the lifted vertices; it is the notion of Definition 3.1 when the positions are distinct, as the positions we use are (`AboveHalf.liftPos_injective`).

The lifts in $A \cup B$ form a graph $\widetilde{X}$ on the lifted vertices, and $(v, i) \mapsto v$ is a two-sheeted covering $\widetilde{X} \to X$: for each edge e at x, the lifted vertex (x, i) lies in exactly one of $\tilde{a}(e)$ and $\sigma(\tilde{a}(e))$. The swap is a deck transformation exchanging A and B, so B is t-degenerate whenever A is, for the order transported by σ. A lift selection is the same as a voltage assignment in $\mathbb{Z}/2$ together with a choice, for each edge, of which of its two lifts in the double cover is called A.

Theorem 3.2 (Theorem D)**.** Lean (partial): `AboveHalf.exists_liftSelection_of_fluid`, `AboveHalf.liftPos_injective` [`AboveHalfRate/LiftSelection.lean`]

Not formalized: the probabilistic existence of labels with small loads (Lemma 3.6), which uses Bernstein's inequality.

Let X be a finite simple graph on n vertices, regular of degree $\delta \geq 1$. Put $\tau = 2 - \sqrt{3}$ and $\nu = \tau(1 - \tau)\delta$, and let $\lambda > 0$ satisfy

$$2n \exp\Big(-\frac{\lambda^2}{2(\nu + \lambda/3)}\Big) < 1. \tag{2}$$

Then X has a lift selection that is t-degenerate for $t = \lceil \tau\delta + \lambda \rceil$. The order can be taken as follows, for injective labels $\xi\colon X^0 \to (0, 1)$: the lifted vertex $(v, 0)$ has position ξ_v, the lifted vertex $(v, 1)$ has position $2 - \xi_v$, and lifted vertices are ordered by position.

Outline of the formal proof. The formal part is `AboveHalf.exists_liftSelection_of_fluid`. From injective labels in $(0,1)$ whose fluid loads (Section 3.3) are at most an integer t, it produces a lift selection A such that A is t-degenerate for the order by position and $\sigma(A)$ for its transport by σ. It rounds the fluid rule by Hall's theorem (Lemma 3.4) and realizes the chosen options by lifts (Lemma 3.3); `AboveHalf.liftPos_injective` says that the positions are distinct. The existence of labels with loads at most $\lceil \tau\delta + \lambda \rceil$ is the probabilistic step, proved in Section 3.4. □

Since $\lambda = O(\sqrt{\delta \log n} + \log n)$ satisfies (2), the degeneracy is $(2-\sqrt{3})\delta + O(\sqrt{\delta \log n})$ when $\log n = O(\delta)$. The constant $2-\sqrt{3} \approx 0.268$ is what the construction of Section 7 pays: local codes of redundancy t carry the separated structure, and the rate obtained is about $(1-t/\delta)^2$. A naive selection does much worse; Remark 3.8 explains why some cleverness is needed.

3.2 Options and rounding

Fix injective labels $\xi\colon X^0 \to (0,1)$ and order the lifted vertices by position as in Theorem 3.2. All first copies $(v,0)$ come before all second copies $(v,1)$; the first copies are in increasing order of label, the second copies in decreasing order.

Lemma 3.3 (Options)**.** Lean: `AboveHalf.isEarlierEnd_lifts`, `AboveHalf.isOption_mk_iff`, `AboveHalf.isOption_iff_exists_lift`, `AboveHalf.liftOf_options`, `AboveHalf.isEarlierEnd_liftOf`, `AboveHalf.isDegenerateFor_image_liftOf` [`AboveHalfRate/LiftSelection.lean`]

Let $e = \{x, y\}$ be an edge with $\xi_x < \xi_y$. The earlier endpoints of its four lifts are

$$\{(x,0),(y,0)\} \mapsto (x,0), \qquad \{(x,0),(y,1)\} \mapsto (x,0),$$
$$\{(x,1),(y,0)\} \mapsto (y,0), \qquad \{(x,1),(y,1)\} \mapsto (y,1).$$

Call $O(e) = \{(x,0),(y,0),(y,1)\}$ the options *of e. For every map o that assigns to each edge one of its options, the lift selection*

$$\tilde{a}(e) = \{(x,0),(y,1)\},\ \{(x,1),(y,0)\},\ \{(x,1),(y,1)\} \quad \textit{for} \quad o(e) = (x,0),\ (y,0),\ (y,1)$$

has earlier endpoint $o(e)$ on every edge. Hence it is t-degenerate if o chooses every lifted vertex for at most t edges.

Outline of the formal proof. The positions satisfy $\xi_x < \xi_y < 1 < 2-\xi_y < 2-\xi_x$, which gives the four earlier endpoints. The construction realizes each option by a lift whose earlier endpoint is that option. □

A *fractional assignment* gives each edge e nonnegative weights $w(e,z)$ on its options $z \in O(e)$, with sum 1. The *load* of a lifted vertex z is $\sum_e w(e,z)$, the sum over the edges having z as an option.

Lemma 3.4 (Rounding)**.** Lean: `AboveHalf.exists_assignment_of_fractional` [`AboveHalfRate/LiftSelection.lean`]

If a fractional assignment has all loads at most an integer T, then some assignment of options has every lifted vertex chosen by at most T edges.

Outline of the formal proof. Make T copies of each lifted vertex and join each edge to the copies of its options. For a set S of edges, the total weight $|S|$ is spread over the options of S, each receiving at most T, so the copies of those options number at least $|S|$. By Hall's theorem [18] the edges can be matched to distinct copies. □

3.3 The fluid rule

Put $\tau = 2 - \sqrt{3}$, the smaller root of $\tau^2 - 4\tau + 1 = 0$; equivalently $(1-\tau)^2 = 2\tau$. For an edge $\{x, y\}$ with $\xi_x < \xi_y$, write $\eta = \xi_y$ and send

$$\min\Big(1, \frac{\tau}{\eta}\Big) \text{ to } (y, 1), \qquad \Big(1 - \frac{\tau}{\eta}\Big)_+ \frac{\eta - \tau}{2\eta} \text{ to } (y, 0), \qquad \Big(1 - \frac{\tau}{\eta}\Big)_+ \frac{\eta + \tau}{2\eta} \text{ to } (x, 0).$$

The weights are nonnegative and sum to 1. They depend only on the larger label. For a vertex z with label u and a neighbour with label $s \neq u$, the weight that the edge between them sends to $(z, 1)$ and to $(z, 0)$ is

$$f_1(u, s) = \begin{cases} \min(1, \tau/u) & s < u, \\ 0 & s > u, \end{cases} \qquad f_0(u, s) = \begin{cases} (1 - \tau/u)_+ \dfrac{u - \tau}{2u} & s < u, \\ (1 - \tau/s)_+ \dfrac{s + \tau}{2s} & s > u, \end{cases}$$

so the load of (z, i) is $\sum_{w \sim z} f_i(u, \xi_w)$.

Lemma 3.5 (Fluid loads)**.** Lean: `AboveHalf.Fluid.charges_sum`, `AboveHalf.Fluid.f1_nonneg`, `AboveHalf.Fluid.f1_le_one`, `AboveHalf.Fluid.f0_nonneg`, `AboveHalf.Fluid.f0_le_half`, `AboveHalf.Fluid.integral_f1`, `AboveHalf.Fluid.integral_f0`, `AboveHalf.Fluid.one_sub_tau_sq`, `AboveHalf.Fluid.integral_f1_sq_sub_le`, `AboveHalf.Fluid.integral_f0_sq_le`, `AboveHalf.Fluid.integral_f0_sq_sub_le` [`AboveHalfRate/Fluid.lean`]

For every $u \in (0, 1]$:

(a) $0 \le f_1(u, s) \le 1$ *and* $0 \le f_0(u, s) \le \frac{1}{2}$ *for* $s \in (0, 1]$*;*

(b) $\int_0^1 f_1(u, s)\, ds = \min(u, \tau) \le \tau$ *and* $\int_0^1 f_0(u, s)\, ds = \frac{(1-\tau)^2}{2} = \tau$*;*

(c) $\int_0^1 f_1(u, s)^2\, ds - \big(\int_0^1 f_1(u, s)\, ds\big)^2 \le \tau(1 - \tau)$ *and* $\int_0^1 f_0(u, s)^2\, ds \le \tau/2 \le \tau(1 - \tau)$.

Outline of the formal proof. (a) is direct. (b) For f_1 the integral is $u \min(1, \tau/u)$. For f_0 it is $(u - \tau)_+^2/(2u) + \frac{1}{2}\int_{\max(u,\tau)}^1 (1 - \tau^2/s^2)\, ds$. When $u \le \tau$ this is $\frac{1}{2}(1 - \tau) - \frac{1}{2}\tau(1 - \tau) = \frac{1}{2}(1 - \tau)^2$. When $u > \tau$ it is $\big((u - \tau)^2 + u(1 - u) - \tau^2(1 - u)\big)/(2u) = \frac{1}{2}(1 - \tau)^2$. Finally $(1 - \tau)^2 = 2\tau$. (c) For $u \le \tau$ the first variance is $u(1 - u) \le \tau(1 - \tau)$; for $u > \tau$ it is $\tau^2(1 - u)/u \le \tau(1 - \tau)$. The second bound follows from $f_0 \le \frac{1}{2}$ and (b). □

So each lifted vertex receives load τ per unit degree in expectation, or less, whatever its label: the rule is balanced exactly at $\tau = 2 - \sqrt{3}$.

3.4 Concentration and the proof of Theorem D

We use Bernstein's inequality [3, Section 2.8]: if $Y_1, \dots, Y_\delta$ are independent random variables with $Y_i - \mathbb{E}Y_i \le M$ and $\sum_i \operatorname{Var} Y_i \le \nu$, then $\Pr\big(\sum_i (Y_i - \mathbb{E}Y_i) \ge \lambda\big) \le \exp\big(-\lambda^2/(2(\nu + M\lambda/3))\big)$ for $\lambda > 0$.

Lemma 3.6 (Concentration)**.** Not formalized; a complete proof is given below.

Let the labels ξ_v be independent and uniform on $(0, 1)$. For every lifted vertex (z, i) and every $\lambda > 0$, the load of (z, i) under the fluid rule is at least $\tau\delta + \lambda$ with probability at most $\exp\big(-\lambda^2/(2(\nu + \lambda/3))\big)$, where $\nu = \tau(1 - \tau)\delta$.

Proof (not formalized). Condition on $\xi_z = u$. Since X is simple, the δ neighbours w of z are distinct vertices other than z, so their labels are independent and uniform on $(0,1)$ given ξ_z. The load is $\sum_{w\sim z} f_i(u,\xi_w)$, a sum of δ independent terms with values in $[0,1]$. By Lemma 3.5(b) its conditional mean is at most $\tau\delta$, and by Lemma 3.5(c) the conditional variances sum to at most ν. Bernstein's inequality with $M = 1$ bounds the conditional probability that the load exceeds its mean by λ, hence the conditional probability that it is at least $\tau\delta + \lambda$, by $\exp(-\lambda^2/(2(\nu + \lambda/3)))$. Integrating over u gives the claim. □

Proof of Theorem 3.2. Choose the labels independently and uniformly on $(0,1)$. They are injective with probability 1, and by Lemma 3.6 and a union bound over the $2n$ lifted vertices, all loads are below $\tau\delta + \lambda$ with probability at least $1 - 2n\exp(-\lambda^2/(2(\nu + \lambda/3))) > 0$. Fix injective labels with this property. The fluid rule is a fractional assignment with all loads at most the integer $t = \lceil \tau\delta + \lambda \rceil$. By Lemma 3.4 some assignment of options has every lifted vertex chosen by at most t edges, and by Lemma 3.3 the corresponding lift selection is t-degenerate for the order by position. □

Only the probabilistic step is not formalized. Given injective labels in $(0,1)$ whose fluid loads are at most t, `AboveHalf.exists_liftSelection_of_fluid` produces a lift selection A such that A and $\sigma(A)$ are t-degenerate for the order by position and its transport by σ (Lemmas 3.3 and 3.4); and the Lean library proves the expected loads and second moments of Lemma 3.5 and the range bounds `AboveHalf.Fluid.f1_sub_le_one` and `AboveHalf.Fluid.f0_sub_le_one`, which are the inputs of Bernstein's inequality.

Corollary 3.7 (The base for $p = 127$)**.** Lean (partial): `AboveHalf.Numbers.base_degeneracy`, `AboveHalf.Numbers.base_bernstein` [`AboveHalfRate/Numbers.lean`]
Not formalized: the application of Theorem 3.2, whose probabilistic step is not formalized.

Every simple 16130*-regular graph on* $n = 2\,197\,862$ *vertices has a* 4640*-degenerate lift selection.*

Outline of the formal proof. Take $\delta = 16130$ and $\lambda = 317$ in Theorem 3.2. Since $\tau(1-\tau) = 3\sqrt{3} - 5 < 0.1961527$, we have $\nu < 3163.95$ and $\lambda^2/(2(\nu + \lambda/3)) > 100489/6539.24 > 15.367$, while $\log(2n) = \log 4\,395\,724 < 15.297$. So (2) holds, and $t = \lceil 16130\tau + 317 \rceil = \lceil 4639.02\ldots \rceil = 4640$. The Lean theorems `AboveHalf.Numbers.base_degeneracy` and `AboveHalf.Numbers.base_bernstein` check (2) and $t = 4640 \le 4663$ exactly, through their own rational bounds on $\sqrt{3}$, on e and on the exponent; the decimals above are numerical evaluations. □

Remark 3.8 (Why the order must be deep)**.** Section 7 needs selections whose degeneracy is a fraction of δ below $1 - 1/\sqrt{2} \approx 0.293$. Orders with only two levels cannot give that. Suppose every lift in A has an endpoint with at most t lifts of A (an *A-light* vertex), every lift in B an endpoint with at most t lifts of B (a *B-light* vertex), and $t < \delta/2$; then the private checks of Section 4 could all be single local checks. The swap exchanges A-light and B-light vertices, so there are equally many, say ℓ. A B-light vertex carries at least $\delta - t > t$ lifts of A, and each has its other end A-light; an A-light vertex carries at most t. Hence $\ell(\delta - t) \le \ell t$, so $\ell = 0$, which is impossible because A is not empty. So $t \ge \delta/2$, and the rate $(1 - t/\delta)^2$ of Section 7 would be at most $1/4$. Deep orders, as in Theorem 3.2, are what make a smaller degeneracy possible, and they are why the private checks have large support. This remark is not used in any proof.

4 Private checks and separated structures

Let G be a locally finite graph with a local datum over a field $\mathbb{F}$ whose local checks interpolate on t coordinates (Section 2.3). A *private check* of an edge a in a set $\widetilde{A}$ of edges is an element $h \in D(G)$

with $h(a) = 1$ and $h(f) = 0$ for every other edge $f \in \widetilde{A}$. Private checks are recovery rules: on the Tanner code, the symbol at a is a linear function of the symbols outside $\widetilde{A}$ (Lemma 2.3).

4.1 Peeling

A *rank function* for a set $\widetilde{A}$ of edges is a map $r\colon G^0 \to \mathbb{N}$ such that the two ends of every edge of $\widetilde{A}$ have different ranks. An edge of $\widetilde{A}$ at w goes *up* from w if its other end has larger rank. A *step down* goes from w to a neighbour of smaller rank along an edge of $\widetilde{A}$, and the *down-set* of u consists of the vertices reachable from u by steps down.

Lemma 4.1 (Peeling)**.** Lean: `AboveHalf.exists_private_check`, `AboveHalf.support_downChecks`, `AboveHalf.exists_walk_of_down` [`AboveHalfRate/Peeling.lean`]

Suppose every vertex has at most t edges of $\widetilde{A}$ going up. Then every edge $a = \{u, v\} \in \widetilde{A}$ with $r(u) < r(v)$ has a private check in $\widetilde{A}$ that is a sum of local checks at vertices of the down-set of u, and every vertex of the down-set is joined to u by a walk of length at most $r(u)$.

Outline of the formal proof. By strong induction on the rank, every edge $\{w, y\} \in \widetilde{A}$ going up from w and every value $c \in \mathbb{F}$ admit a sum h of local checks at the down-set of w with $h(\{w, y\}) = c$ and $h = 0$ on the rest of $\widetilde{A}$. At most t edges of $\widetilde{A}$ go up from w, so a local check g_w at w takes the value c on $\{w, y\}$ and 0 on the other edges going up. For each edge $\{w, z\} \in \widetilde{A}$ going down from w, the induction hypothesis at z, with value $-g_w(\{w, z\})$, gives a correction h_z at the down-set of z, which lies in the down-set of w. Then $h = g_w + \sum_z h_z$ has the required values. On the edges at w this holds because no edge of $\widetilde{A}$ at w joins vertices of equal rank. Away from w it holds because each h_z vanishes on $\widetilde{A}$ outside its own edge $\{w, z\}$. Each step down lowers the rank by at least 1, which bounds the walks. □

A sum of local checks at the down-set of u is supported in the stars of its vertices, so by the definition of the distance between edges (Section 2.2) every edge f in the support of the private check of a satisfies $\operatorname{dist}(a, f) \le r(u) + 1$. This step uses the line-graph metric, which is not formalized.

The recursion explores the whole down-set, so the support of a private check is controlled only by the depth of the rank function, not by its degeneracy alone. The checks are explicit given the local checks and the ranks.

4.2 Separated structures

From now on a group Γ acts on G freely on vertices and without inversions, hence freely on edges, and the local datum is Γ-equivariant.

Definition 4.2 (Twisted separated structure)**.** A *twisted separated structure* of degeneracy t and depth R consists of a homomorphism $\chi\colon \Gamma \to \mathbb{Z}/2$, a partition $G^1 = \widetilde{A} \sqcup \widetilde{B}$ with

$$\gamma\widetilde{A} = \widetilde{A} \text{ if } \chi(\gamma) = 0, \qquad \gamma\widetilde{A} = \widetilde{B} \text{ if } \chi(\gamma) = 1,$$

and rank functions $r_A, r_B\colon G^0 \to \{0, \dots, R\}$ for $\widetilde{A}$ and for $\widetilde{B}$ such that every vertex has at most t edges of $\widetilde{A}$ going up for r_A, and at most t edges of $\widetilde{B}$ going up for r_B. The *class* of an edge is $\operatorname{cls}(e) = 0$ on $\widetilde{A}$ and 1 on $\widetilde{B}$, so that $\operatorname{cls}(\gamma e) = \operatorname{cls}(e) + \chi(\gamma)$.

Proposition 4.3 (Separated checks)**.** Not formalized; a complete proof is given below.

Suppose G carries a twisted separated structure of degeneracy t and depth R, and the local checks interpolate on t coordinates. Then there is a family $(h_e)_{e \in G^1}$ in $D(G)$ such that for all $e \in G^1$ and $\gamma \in \Gamma$:

(a) $h_{\gamma e} = \gamma h_e$*;*

(b) $h_e(e) = 1$*, and* $h_e(f) = 0$ *for every edge* $f \neq e$ *of the class of* e*;*

(c) every edge f *in the support of* h_e *satisfies* $\operatorname{dist}(e, f) \leq R$*.*

Proof (not formalized). Choose one edge in every Γ-orbit of edges. For a chosen edge $a = \{u, v\}$ of class c, orient it so that $r(u) < r(v)$ for the rank function r of its class, and let h_a be the private check of Lemma 4.1 in that class. Since $r(u) < r(v) \leq R$, every edge f of its support has $\operatorname{dist}(a, f) \leq r(u) + 1 \leq R$, by the paragraph after Lemma 4.1. Every edge e is γa for exactly one chosen a and exactly one $\gamma \in \Gamma$, because the action on edges is free; put $h_e = \gamma h_a$. It lies in $D(G)$ because the datum is equivariant. Then (a) holds by uniqueness of γ, and (c) because Γ acts by isometries. For (b), $h_e(e) = h_a(a) = 1$. If $f \neq e$ has the class of e, then $\gamma^{-1} f \neq a$ has class $\operatorname{cls}(f) - \chi(\gamma) = \operatorname{cls}(e) - \chi(\gamma) = c$, so $h_e(f) = h_a(\gamma^{-1} f) = 0$. □

Of Proposition 4.3 only Lemma 4.1 is formalized; the transport and the metric bookkeeping are written proofs, and the Lean main theorems take the family of checks as a hypothesis. The family is the input of the descent of Section 5. In Lean it is a `AboveHalf.CheckSystem`, whose fields `check_smul`, `check_self` with `sep`, and `radius` are (a), (b) and (c); its other fields are the class function and the twist of Definition 4.2, related by `cls_smul`, and a metric on the coordinates.

The recovery relation $e \mapsto \operatorname{supp} h_e \setminus \{e\}$ on the edges of G is perfectly separated by the classes, so its incidence graph never joins L_e to R_e. The twist χ is what lets quotients escape the half bound of perfectly separated relations: an element of Γ with $\chi = 1$ exchanges the classes.

4.3 Separated structures from lift selections

Proposition 4.4 (Lifting a selection)**.** Not formalized; a complete proof is given below.

Let $\mathcal{T}$ *be a tree,* Γ *a group acting on it freely on vertices and without inversions with* $s(\Gamma) \geq 3$*, and* $X = \Gamma \backslash \mathcal{T}$*, a finite simple graph on* n *vertices. If* X *has a* t*-degenerate lift selection, then* $\mathcal{T}$ *carries a twisted separated structure of degeneracy* t *and depth* $2n - 1$*.*

Proof (not formalized). Let A be the selection, $B = \sigma(A)$, and $\widetilde{X}$ the double cover of Section 3.1, with covering maps $\pi \colon \mathcal{T} \to X$ and $\varpi_0 \colon \widetilde{X} \to X$. Number the $2n$ lifted vertices $0, \dots, 2n - 1$ in the order of degeneracy. Fix a vertex x_0 of $\mathcal{T}$ and a lift of $\pi(x_0)$ to $\widetilde{X}$, and define $\varpi \colon \mathcal{T} \to \widetilde{X}$ by lifting along ϖ_0 the images under π of the unique paths from x_0 (Section 2.2). Then $\varpi_0 \circ \varpi = \pi$, and ϖ maps the edges at each vertex bijectively onto the edges at its image, since π and ϖ_0 do.

For $\gamma \in \Gamma$, $\varpi \circ \gamma$ also lifts π, because $\pi \circ \gamma = \pi$. Two lifts of a map from a connected graph agree everywhere if they agree at one vertex, and a vertex of X has two lifts; so $\varpi \circ \gamma = \varpi$ or $\varpi \circ \gamma = \sigma \circ \varpi$. Put $\chi(\gamma) = 0$ or 1 accordingly. Since $\sigma^2 = 1$ this is a homomorphism. Let $\widetilde{A} = \varpi^{-1}(A)$ and $\widetilde{B} = \varpi^{-1}(B)$, a partition of the edges of $\mathcal{T}$, with $\gamma \widetilde{A} = \widetilde{A}$ if $\chi(\gamma) = 0$ and $\gamma \widetilde{A} = \varpi^{-1}(\sigma A) = \widetilde{B}$ if $\chi(\gamma) = 1$.

Let $r_A(x)$ be the number of $\varpi(x)$ and $r_B(x)$ that of $\sigma(\varpi(x))$. The edges of $\widetilde{A}$ at x correspond bijectively to the lifts of A at $\varpi(x)$, with the same numbers at their other ends, so their ends have different ranks and at most t of them go up. The same holds for $\widetilde{B}$, with $\sigma \circ \varpi$ in place of ϖ, since $B = \sigma(A)$ is t-degenerate for the order transported by σ. □

Combined with Theorem 3.2, Proposition 4.4 turns every quotient of a regular tree by a group acting freely and without inversions, with simple quotient, into the input of Proposition 4.3. Depth $2n - 1$ is what makes the incidence degree bound of Section 7 astronomically large: the private checks of Lemma 4.1 may reach every rank.

5 Twisted quotients

5.1 Descent of separated checks

Keep the setting of Proposition 4.3: a family $(h_e)_{e\in G^1}$ in $D(G)$ satisfying (a)–(c) there. Let $\Gamma' \subseteq \Gamma$ be a subgroup with finite quotient $X' = \Gamma'\backslash G$ and projection π, and let $\beta = \sup_e |B(e,R)|$, assumed finite. The *descended relation* on the coordinate set $V' = X'^1$ is

$$N'(\pi e) = \pi\big(\operatorname{supp} h_e \setminus \{e\}\big),$$

which does not depend on the choice of e in its orbit, by (a). A word $c\colon V' \to \mathbb{F}$ is *annihilated by the descended checks* if $\sum_f h_e(f)\, c(\pi f) = 0$ for every $e \in G^1$.

Proposition 5.1 (Descent)**.** Lean: `AboveHalf.CheckSystem.qrel`, `AboveHalf.CheckSystem.isStorageCode_qrel`, `AboveHalf.CheckSystem.card_qrel_le`, `AboveHalf.CheckSystem.card_in_qrel_le`, `AboveHalf.CheckSystem.le_mul_of_edist_le` [`AboveHalfRate/Descent.lean`]

(a) Every set of words annihilated by the descended checks is a storage code for N'.

(b) Both incidence degrees of N' are at most $\beta - 1$.

(c) If every element of $\Gamma' \setminus \{1\}$ moves every edge by at least s, then every coordinate of N' has paired distance at least s/R.

Outline of the formal proof. (a) For $e \in G^1$, split the relation $\sum_f h_e(f)c(\pi f) = 0$ into the term $f = e$, which is $c(\pi e)$ because $h_e(e) = 1$, and the other terms, whose coordinates πf lie in $N'(\pi e)$. So two annihilated words that agree on $N'(\pi e)$ agree at πe.

(b) The recovery set $N'(\pi e)$ is the image of $\operatorname{supp} h_e \setminus \{e\} \subseteq B(e,R) \setminus \{e\}$. For the other degree fix $p \in G^1$. Every recovery set $N'(\pi e)$ containing πp has a representative e with $p \in \operatorname{supp} h_e$ and $e \neq p$: if $\pi y = \pi p$ with $y \in \operatorname{supp} h_{e_0} \setminus \{e_0\}$, translate e_0 by the element of Γ' that carries y to p. Different recovery sets get different representatives, all in $B(p,R) \setminus \{p\}$.

(c) Give a port L_f the signed class $\operatorname{cls}(f)$ and a port R_f the signed class $\operatorname{cls}(f) + 1$. An incidence edge $L_y R_e$ upstairs has $y \in \operatorname{supp} h_e$ and $y \neq e$, so y and e have different classes by (b) of Proposition 4.3, and the two ports have the same signed class; they are within distance R by (c). Every incidence edge of N' lifts to such an edge at a prescribed lift of either end, as in the proof of (b). So a path of length k from $L_{\pi a}$ to $R_{\pi a}$ lifts to a path from L_a to $R_{\gamma a}$ with $\gamma \in \Gamma'$, along which the signed class is constant and the coordinate moves by at most kR. Constancy gives $\operatorname{cls}(a) = \operatorname{cls}(\gamma a) + 1 = \operatorname{cls}(a) + \chi(\gamma) + 1$, so $\chi(\gamma) = 1$ and $\gamma \neq 1$. Hence $s \leq \operatorname{dist}(a, \gamma a) \leq kR$. □

Part (a) does not use the displacement at all. Neither does (b). Without displacement, however, recovery sets may contain their own coordinate and paired distances may be short. Part (c) is where the twist pays: a paired path must travel from a coordinate to a translate of it in the other class, and translates by Γ' are far apart.

Lemma 5.2 (Validity on quotients)**.** Not formalized; a complete proof is given below.

Every word of the Tanner code $T(X')$ is annihilated by the descended checks.

Proof (not formalized). For $c \in T(X')$ the function $c \circ \pi$ lies in $T(G)$, and $\sum_f h_e(f)c(\pi f) = \langle h_e, c \circ \pi\rangle = 0$ by Lemma 2.3, since $h_e \in D(G)$. The last step is `AboveHalf.pairing_eq_zero_of_mem_iSup`; the Lean library has no quotient Tanner codes, and its main theorems take the annihilation as a hypothesis. □

5.2 The combinatorial core

Theorem 5.3 (Theorem T)**.** Lean (partial): `AboveHalf.not_Q91At_of_tower`, `AboveHalf.not_Q91BinaryAt_of_tower` [`AboveHalfRate/Main.lean`]
Not formalized: the passage from a twisted separated structure to the family of separated checks (Proposition 4.3) and the facts about the line-graph metric used there, and the annihilation of the codes $T(X_j)$ by the descended checks (Lemma 5.2); the Lean theorems take the family of checks, with its radius and ball sizes, and the annihilation as hypotheses.

Let G be a locally finite graph and Γ a group acting on it freely on vertices and without inversions, with a Γ-equivariant local datum over a finite field $\mathbb{F}$ whose local checks interpolate on t coordinates. Suppose G carries a twisted separated structure of degeneracy t and depth $R \geq 1$, and let $\beta = \sup_e |B(e,R)| < \infty$. Let $(\Gamma_j)_{j\geq 0}$ be subgroups of Γ with finite quotients $X_j = \Gamma_j \backslash G$, such that $|X_j^1| \to \infty$, the displacement $s(\Gamma_j) \to \infty$, and $\dim_{\mathbb{F}} T(X_j) \geq \rho |X_j^1|$ for all j.

(a) The descended relations N_j on X_j^1 have incidence degree at most $\beta - 1$, every coordinate has paired distance at least $s(\Gamma_j)/R$, and $T(X_j)$ is an $\mathbb{F}$-linear storage code for N_j of rate at least ρ.

(b) If $\rho > 1/2$, Question 2.1 has a negative answer at the degree bound $\Delta = \beta - 1$, with the fixed alphabet $\mathbb{F}$.

(c) If moreover $\iota \colon \mathbb{F} \to \{0,1\}^b$ is injective and $\rho \log_2 |\mathbb{F}|/b > 1/2$, Question 2.1 has a negative answer for binary codes at the degree bound $b(\beta - 1)$.

Outline of the formal proof. Proposition 4.3 gives the family of checks. Part (a) is Proposition 5.1 and Lemma 5.2; the rate of $T(X_j)$ over $\mathbb{F}$ is $\dim T(X_j)/|X_j^1|$. The paired-distance condition holds in the strongest form: for every radius K, once $s(\Gamma_j) > RK$ no coordinate has paired distance at most K. In (b) the codes $T(X_j)$ violate the conclusion of Question 2.1. In (c), the binary blow-up of Section 8 multiplies degrees by at most b, does not shorten paired distances, and multiplies rates by $\log_2 |\mathbb{F}|/b$, so its rate is at least $\rho \log_2 |\mathbb{F}|/b$. The Lean statements conclude $\neg$ `MergeDepth.Q91At`$(\beta - 1)$ and $\neg$ `MergeDepth.Q91BinaryAt`$(b(\beta - 1))$ from a family of checks satisfying (a)–(c) of Proposition 4.3, balls of size at most β, a tower of subgroups with finite quotients of size tending to infinity and displacement tending to infinity, and codes on the quotients annihilated by the descended checks with dimension at least ρ times their length. □

Theorem T separates the two halves of the problem. The combinatorial half asks for an equivariant MDS datum on a tree with a twisted separated structure. The algebraic half asks for quotients whose Tanner codes are much larger than counting predicts: dimension counting alone gives $\dim T(X_j) \geq (1 - 2t/(q+1))|X_j^1|$ for local codes of redundancy t on a $(q+1)$-regular graph, which is less than $1/2$ whenever t is above a quarter of the degree. The next section supplies quotients whose Tanner codes have rate at least about $(1 - t/(q+1))^2$, the rate of a tensor product of two local codes.

6 Twisted Goren–Oort Tanner codes

This section is not formalized. It supplies the algebraic half of Theorem T: an equivariant datum with MDS local codes on the Bruhat–Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_{p^2})$, and a tower of congruence quotients whose Tanner codes have rate close to the square of the local rate (Theorem 6.9). The twelfth paper of the series proves the case $p = 3$ in weight 2 in detail [6, Section 6], from the same sources, and we follow it. Three steps are new: the Riemann–Roch estimate in weight k, where the Hasse invariant

gives the vanishing (Lemma 6.3); the girth bound (Lemma 6.2); and rigidity (Lemma 6.7), which uses the group $\mathrm{SL}_2(\mathbb{Q}_{p^2})$ rather than the icosian group, because for $p > 3$ the icosian group has many orbits on the edges of the tree.

6.1 The icosian group at an inert prime

Let $E = \mathbb{Q}(\sqrt{5})$, with ring of integers $\mathcal{O}_E = \mathbb{Z}[\theta]$, $\theta = (1+\sqrt{5})/2$; it has narrow class number one, θ has norm -1, and every totally positive unit is a square. Let B be the quaternion algebra over E ramified exactly at the two infinite places, and $I \subset B$ the icosian ring, a maximal order of class number one and type number one whose norm-one units form the binary icosahedral group $2I \cong \mathrm{SL}_2(\mathbb{F}_5)$ of order 120 [40, Proposition V.3.4, Corollaire I.4.11 and p. 155].

Fix a prime $p \geq 7$ inert in E, that is, $p \equiv \pm 2 \pmod 5$. The completion of E at p is the unramified quadratic extension $\mathbb{Q}_{p^2}$ of $\mathbb{Q}_p$, with ring of integers $\mathbb{Z}_{p^2}$ and residue field $\mathbb{F}_{p^2}$, and B splits there: $B \otimes_E \mathbb{Q}_{p^2} \cong M_2(\mathbb{Q}_{p^2})$ with $I \otimes \mathbb{Z}_{p^2} \cong M_2(\mathbb{Z}_{p^2})$. The group

$$\Gamma^+ = I[1/p]^1/\{\pm 1\} \subset \mathrm{PSL}_2(\mathbb{Q}_{p^2})$$

acts on the Bruhat–Tits tree $\mathcal{T}$ of $\mathrm{PGL}_2(\mathbb{Q}_{p^2})$. Its vertices are the homothety classes of $\mathbb{Z}_{p^2}$-lattices in $\mathbb{Q}_{p^2}^2$; the neighbours of $[\Lambda]$ are the classes of the lattices Λ' with $p\Lambda \subset \Lambda' \subset \Lambda$ and $\Lambda/\Lambda' \cong \mathbb{F}_{p^2}$, so the edges at $[\Lambda]$ correspond to the points of $\mathbb{P}^1(\Lambda/p\Lambda)$, and $\mathcal{T}$ is (p^2+1)-regular. The vertices have two parities, the parity of $\log_p|\det|$ of a basis, which elements of determinant one preserve. Let Λ_0 be the vertex of $I \otimes \mathbb{Z}_{p^2}$.

Lemma 6.1 (The icosian group and its congruence subgroups)**.** Not formalized; a complete proof is given below.

(a) There is $\alpha_0 \in I$ of reduced norm p, and $\Lambda_1 = \alpha_0\Lambda_0$ is a neighbour of Λ_0. The group Γ^+ acts on $\mathcal{T}$ without inversions and transitively on the vertices of each parity; the stabilizers of Λ_0 and Λ_1 are $A_5 = 2I/\{\pm 1\}$ and $\alpha_0 A_5 \alpha_0^{-1}$.

(b) Let λ_0 be a prime of $\mathcal{O}_E$ of prime norm $\ell \geq 7$, and for $j \geq 1$ let $\Gamma^+(\lambda_0^j) = \ker(\Gamma^+ \to \mathrm{PSL}_2(\mathcal{O}_E/\lambda_0^j))$. The map is onto, $\Gamma^+(\lambda_0^j)$ is torsion-free and acts freely on $\mathcal{T}$, the groups $\Gamma^+(\lambda_0^j)$ strictly decrease with trivial intersection, and $X_j = \Gamma^+(\lambda_0^j)\backslash\mathcal{T}$ has

$$c_j = |\mathrm{PSL}_2(\mathcal{O}_E/\lambda_0^j)|/60$$

vertices of each parity.

Proof (not formalized). (a) This is the argument of [6, Lemma 6.2] with 3 replaced by p. A vertex is the class of the completion at p of a right I-ideal $\mathfrak{a}$ with $\mathfrak{a}_\mathfrak{q} = I_\mathfrak{q}$ for $\mathfrak{q} \nmid p$, and $\mathfrak{a} = \beta I$ with $\beta \in I[1/p]^\times$ because I has class number one. The reduced norm of β is a totally positive unit of $\mathcal{O}_E[1/p]$, hence $p^a\theta^{2b}$, and we may take $b = 0$. Applied to a neighbour Λ' of Λ_0 with $\Lambda' \subset \Lambda_0$, whose ideal has reduced norm the prime ideal (p), this gives α_0. For a general vertex, $\beta/p^{a/2}$ (for even a) or $(\beta/p^{(a-1)/2})\alpha_0^{-1}$ (for odd a) lies in $I[1/p]^1$ and moves Λ_0 or Λ_1 to it. Elements of reduced norm one preserve parity, so there are no inversions. The stabilizer of Λ_0 in $I[1/p]^1$ consists of the elements that are integral at p, which is $I^1 = 2I$; conjugation by $\alpha_0 \in I[1/p]^\times$ preserves $I[1/p]^1$ and gives the stabilizer of Λ_1.

(b) This is [6, Lemma 6.3] with 3 and 11 replaced by p and ℓ. Strong approximation for the simply connected group B^1, which is not compact at p [31, Section 7.4], gives surjectivity. A nontrivial element of finite order of Γ^+ fixes a vertex [34, Section I.4], so it is conjugate into A_5 or $\alpha_0 A_5 \alpha_0^{-1}$,

which map injectively to $\mathrm{PSL}_2(\mathcal{O}_E/\lambda_0)$ because $1+\lambda_0 M_2(\mathcal{O}_{E,\lambda_0})$ is torsion-free for odd ℓ. Hence $\Gamma^+(\lambda_0^j)$ is torsion-free and, acting without inversions, acts freely. It is normal and meets no vertex stabilizer, so the vertices of X_j of a given parity form the double coset space $\Gamma^+(\lambda_0^j)\backslash\Gamma^+/A_5$, of size $|\mathrm{PSL}_2(\mathcal{O}_E/\lambda_0^j)|/60$, and likewise for the other parity. The decrease and the trivial intersection are proved as in [6, Lemma 6.3]. □

So X_j is bipartite, with c_j vertices of each parity and $(p^2+1)c_j$ edges, and it is (p^2+1)-regular because $s(\Gamma^+(\lambda_0^j)) \geq 2$ (Lemma 6.2 below, with $r=0$, and Section 2.2); it is the dual graph of the Goren–Oort divisor below. Its girth grows along the tower, by an archimedean argument in the style of Lubotzky, Phillips and Sarnak [27].

Lemma 6.2 (Girth)**.** Not formalized; a complete proof is given below.

Let $j \geq 1$ and $r \geq 0$ with $4p^r < \ell^j$. Then every element of $\Gamma^+(\lambda_0^j) \setminus \{1\}$ moves every vertex of $\mathcal{T}$ by at least $2r+2$. Hence $s(\Gamma^+(\lambda_0^j)) \geq 2r+2$; in particular X_1 is simple when $\ell > 4p$, and $s(\Gamma^+(\lambda_0^j)) \geq 2(j\log\ell - \log 4)/\log p$ for all j.

Proof (not formalized). Elements of Γ^+ preserve parity, so they move vertices by even distances. Suppose $\gamma \neq 1$ in $\Gamma^+(\lambda_0^j)$ moves a vertex x by $2r'$ with $1 \leq r' \leq r$. By Lemma 6.1(a), $x = g\Lambda_0$ or $x = g\alpha_0\Lambda_0$ with $g \in \Gamma^+$; conjugating by g and by α_0, which preserve $\Gamma^+(\lambda_0^j)$ (the first because it is normal, the second because α_0 is a unit at λ_0), we may assume $x = \Lambda_0$. Let $\tilde{\gamma} \in I[1/p]^1$ be the lift with $\tilde{\gamma} \equiv 1 \pmod{\lambda_0^j}$; it is unique because $-1 \not\equiv 1 \pmod{\lambda_0}$ [6, Lemma 6.3]. By the Cartan decomposition, $\alpha = p^{r'}\tilde{\gamma}$ lies in $M_2(\mathbb{Z}_{p^2}) = I \otimes \mathbb{Z}_{p^2}$, hence in I, and $\alpha \equiv p^{r'} \pmod{\lambda_0^j I}$. Let π be a totally positive generator of λ_0^j and write $\alpha = p^{r'} + \pi\beta$ with $\beta \in I$. Then $\beta \neq 0$ because $\gamma \neq 1$, and

$$p^{2r'} = \mathrm{Nrd}\,\alpha = p^{2r'} + p^{r'}\pi\,\mathrm{Tr}\,\beta + \pi^2\,\mathrm{Nrd}\,\beta$$

gives $p^{r'}\,\mathrm{Tr}\,\beta = -\pi\,\mathrm{Nrd}\,\beta$. Since π is prime to p, $\mathrm{Tr}\,\beta = \pi u$ with $u \in \mathcal{O}_E$, and then $\mathrm{Nrd}\,\beta = -p^{r'}u$, so $u \neq 0$. At each real embedding ι of E, $B \otimes_\iota \mathbb{R}$ is the algebra of Hamilton quaternions, so $\iota(\mathrm{Tr}\,\beta)^2 \leq 4\,\iota(\mathrm{Nrd}\,\beta)$, that is, $\iota(\pi)^2\iota(u)^2 \leq 4p^{r'}|\iota(u)|$, hence $\iota(\pi)^2|\iota(u)| \leq 4p^{r'}$. Multiplying over the two embeddings gives $\ell^{2j}|N(u)| \leq 16p^{2r'}$, and $|N(u)| \geq 1$, so $\ell^j \leq 4p^{r'} \leq 4p^r$, a contradiction. The displacement bound follows from (1). For $j=1$ and $r=1$ the quotient is simple by Section 2.2. The last bound takes the largest admissible r. □

6.2 Riemann–Roch in weight k

Let $\mathfrak{n} = \lambda_0^j$, an ideal prime to p with a totally positive generator, and $\mathcal{K}(\mathfrak{n}) \subset \mathrm{GL}_2(\hat{\mathcal{O}}_E)$ its principal congruence subgroup. It is neat: a root of unity congruent to 1 modulo a prime above ℓ has ℓ-power order, so it is 1 or has degree at least $\ell - 1 \geq 6$ over $\mathbb{Q}$, while eigenvalues of elements of $\mathrm{GL}_2(E)$ have degree at most 4. As in [6, Section 6.2], choose a smooth projective toroidal compactification over $W(\bar{\mathbb{F}}_p)$ of the closure of one geometric connected component of the Hilbert modular surface of E of level $\mathcal{K}(\mathfrak{n})$, and let S be its special fibre, with boundary D. We take the one given at every cusp by Hirzebruch's minimal cone decomposition [20, Section 2.5]. Every cusp has type $(\mathcal{O}_E, \mathcal{U}_\mathfrak{n})$ up to a totally positive scalar [6, Section 6.2, (G4)], and $\theta^2 = 3 - \theta^{-2}$ has the purely periodic continued fraction $[[3]]$, so the rays of the decomposition are spanned by lattice points A_k ($k \in \mathbb{Z}$) with $A_{k-1} + A_{k+1} = 3A_k$, any two consecutive ones forming a basis of the lattice [20, Sections 2.5 and 2.6]. So the decomposition is smooth. It is also projective in the sense of [25, Definitions 7.3.1.1 and 7.3.1.3]. Let φ be the function that is linear on each cone and equal to 1 at every A_k. It takes positive integer values at nonzero lattice points, by smoothness. If $x = aA_{k-1} + a'A_k$ and $y = c'A_k + cA_{k+1}$ with nonnegative coefficients, then $\varphi(x+y) - \varphi(x) - \varphi(y) = \min(a,c)$, by

$A_{k-1} + A_{k+1} = 3A_k$. So φ bends at every ray, hence it is concave (superadditive, Lan's convexity) and its maximal domains of linearity are exactly the cones: it is a polarization function. The complex fibre of the component is $\Gamma(\mathfrak{n})\backslash\mathfrak{H}^2$ with $\Gamma(\mathfrak{n}) \subset \mathrm{SL}_2(\mathcal{O}_E)$. We use the facts (G1)–(G4) of [6, Section 6.2], whose sources hold for every prime unramified in E and every level prime to it; we restate them for p.

(G1) S is smooth, projective and connected, and $D \neq 0$ is a disjoint union of cycles of smooth rational curves, one for each cusp [32, Theorem 5.1, Proposition 5.2 and Corollary 5.3], [12], [15, Section 7], [25, Theorems 6.4.1.1 and 7.3.3.4].

(G2) The Hodge bundle splits as $\omega_1 \oplus \omega_2$ because p is unramified, and $K_S + D \cong \omega_1^2 \otimes \omega_2^2$ [25, Theorem 6.4.1.1(4)]. Write $H = \omega_1 \otimes \omega_2$ and, for an integer k, $L_k = H^{\otimes k}$, the bundle of parallel weight k; so $K_S + D = 2H$ in additive notation.

(G3) The partial Hasse invariants $h_1 \in H^0(\omega_1^{-1}\omega_2^p)$ and $h_2 \in H^0(\omega_1^p\omega_2^{-1})$ have reduced divisors Z_1, Z_2, regular curves meeting in finitely many points [17, Section 1 and Theorem 2.1(1)]; they extend to S by the Koecher principle [15, Theorem 8.3] and do not vanish on D, because their q-expansions at every cusp are 1 [17, Theorem 2.1(2)]. So $h = h_1h_2 \in H^0(L_{p-1})$ has divisor $Y = Z_1 + Z_2$, and $\mathcal{O}(Y) \cong L_{p-1}$. Since p is inert, Y is the reduced supersingular locus [2, Proposition 4.1].

(G4) Let $e = |\mathrm{PSL}_2(\mathcal{O}_E/\mathfrak{n})| \cdot 2\zeta_E(-1) = |\mathrm{PSL}_2(\mathcal{O}_E/\mathfrak{n})|/15$, the Euler number of $\Gamma(\mathfrak{n})\backslash\mathfrak{H}^2$ [20, Sections 1.6 and 1.7]. Every cusp is resolved by a cycle of curves of self-intersection -3 [20, Sections 2.5 and 2.6], and $K_S^2 = 2e + \sum(2-b) = 2e - \#\{\text{boundary curves}\}$, where the sum runs over the boundary curves, of self-intersection $-b = -3$ [20, Sections 3.2 and 3.6]. Noether's formula $\chi(\mathcal{O}_S) = (K_S^2 + c_2(S))/12$, with $c_2(S) = e + \#\{\text{boundary curves}\}$, then gives $\chi(\mathcal{O}_S) = e/4 = c_j$, as in [6, Section 6.2, (G4)]. The boundary curves over each cusp are the closures of the strata of the rays of the decomposition, which are smooth over $W(\bar{\mathbb{F}}_p)$ [25, Theorem 6.4.1.1(2) and (3)], so the configuration of the boundary is the same on S and on the complex fibre; $\chi(\mathcal{O})$ and intersection numbers of line bundles are constant in the smooth proper family over $W(\bar{\mathbb{F}}_p)$. So these facts may be checked on the complex fibre, where they are Hirzebruch's. Each cycle has at least five curves: its length is the index of the squares of the units congruent to 1 modulo $\mathfrak{n}$ in the totally positive units [6, Section 6.2, (G4)], which is the order of θ in $(\mathcal{O}_E/\mathfrak{n})^\times/\{\pm1\}$, and for $1 \le a \le 4$ the numbers $\theta^a \mp 1$ have norms ±1, ±4, ±5 or 9, which no prime $\ell \ge 7$ divides.

Lemma 6.3 (Sections in weight k)**.** Not formalized; a complete proof is given below.

For $2 \le k \le p-2$, $h^0(S, L_k) \ge (k-1)^2 c_j$, and restriction $H^0(S, L_k) \to H^0(Y, L_k|_Y)$ is injective. Hence $h^0(Y, L_k|_Y) \ge (k-1)^2 c_j$.

Proof (not formalized). Intersection numbers. Each boundary cycle D_i has arithmetic genus 1, and the cycles are disjoint, so adjunction gives $(K_S + D) \cdot D = 0$; hence $H \cdot D = 0$ and $K_S \cdot D = -D^2$. A cycle of curves of self-intersections $-b$ has square $\sum(2-b)$, so $D^2 = -\#\{\text{boundary curves}\}$. Then $(2H)^2 = (K_S + D)^2 = K_S^2 - D^2 = 2e$ by (G4), so $H^2 = e/2 = 2c_j$, and $H \cdot K_S = H \cdot (2H - D) = 4c_j$.

Riemann–Roch. For a line bundle L on the smooth projective surface S, $\chi(L) = \chi(\mathcal{O}_S) + \frac{1}{2}L \cdot (L - K_S)$ [26, Section 4.1]. For $L = L_k = kH$ this is

$$\chi(L_k) = c_j + \tfrac{1}{2}(k^2H^2 - kH \cdot K_S) = c_j + k^2c_j - 2kc_j = (k-1)^2 c_j.$$

Vanishing. By Serre duality $h^2(L_k) = h^0(K_S - L_k) = h^0((2-k)H - D)$. A nonzero section of this bundle, multiplied by a nonzero section of $\mathcal{O}(D)$, gives a nonzero section of $-(k-2)H$ when

$k \geq 3$, and is a nonzero section of $\mathcal{O}(-D)$ when $k = 2$. The latter does not exist because D is a nonzero effective divisor on the integral projective surface S. If $s \neq 0$ is a section of $-aH$ with $a \geq 1$, then $s^{p-1}h^a$ is a nonzero section of $\mathcal{O}_S$, hence a nonzero constant, so h vanishes nowhere and $(p-1)H \cong \mathcal{O}(Y)$ is trivial by (G3). This contradicts $H^2 = 2c_j > 0$. So $h^2(L_k) = 0$ and $h^0(L_k) \geq \chi(L_k) = (k-1)^2 c_j$.

Injectivity. The kernel of restriction to Y is $H^0(L_k(-Y)) = H^0(-(p-1-k)H)$, by (G3), and $p-1-k \geq 1$; it vanishes by the same argument. □

6.3 Uniformization of the supersingular locus

Let $\mathcal{M}$ be the Rapoport–Zink space of supersingular polarized p-divisible groups with $\mathcal{O}_E \otimes \mathbb{Z}_p$-action, J the group of $\mathcal{O}_E$-linear self-quasi-isogenies of its framing object that multiply the polarization by a factor in $\mathbb{Q}_p^\times$, and $\mathcal{M}_{\mathrm{red}}^{(n)}$ the part of its reduced locus described in [6, Section 6.3], after Kudla and Rapoport [24, Section 4] and Stamm [35]. It is a union of projective lines: a line $C_\Lambda = \mathbb{P}^1(\Lambda/p\Lambda)$ of the first kind for each $\mathbb{Z}_{p^2}$-lattice Λ of a two-dimensional $\mathbb{Q}_{p^2}$-space U_0 with $\langle \Lambda, \Lambda \rangle = p^n \mathbb{Z}_{p^2}$, and a line of the second kind for each such lattice of a second space U_1, which a σ-semilinear bijection $\Phi\colon U_1 \to U_0$ commuting with J identifies with U_0. Here $\mathcal{O}_E \otimes \mathbb{Z}_p$ acts on U_0 by scalars, through the embedding into $W = W(\bar{\mathbb{F}}_p)$ that defines the grading of the isocrystal [24, (4.3)]. Lines of the same kind are disjoint; each line meets exactly one line of the other kind at each of its p^2+1 points rational over $\mathbb{F}_{p^2}$, the superspecial points, and no line elsewhere [24, Lemma 4.3(iii), the text after (4.9) and Proposition 4.4]; the crossings are ordinary double points [2, Theorem 6.1]. Sending a line of the first kind to $[\Lambda]$ and one of the second kind to $[\Phi\Lambda']$ identifies the dual graph with the tree $\mathcal{T}_{U_0}$ of U_0, equivariantly for the subgroup J^0 of elements whose similitude factor is a p-adic unit [6, Section 6.3]. The Hodge bundles of the universal p-divisible group, and so the bundles L_k, carry a canonical action of J.

Lemma 6.4 (Uniformization)**.** Not formalized; a complete proof is given below.

There are an integer n, an action of Γ^+ on $\mathcal{M}_{\mathrm{red}}^{(n)}$ through J^0, and a bijection $\Psi\colon U_0 \to \mathbb{Q}_{p^2}^2$, either $\mathbb{Q}_{p^2}$-linear or σ-semilinear, carrying $\mathcal{T}_{U_0}$ to $\mathcal{T}$, such that:

(a) the identification of the dual graph of $\mathcal{M}_{\mathrm{red}}^{(n)}$ with $\mathcal{T}$ is Γ^+-equivariant;

(b) $Y \cong \Gamma^+(\lambda_0^j)\backslash \mathcal{M}_{\mathrm{red}}^{(n)}$, and under this isomorphism $L_k|_Y$ is the descent of L_k with its J-action; in particular the dual graph of Y is X_j;

(c) L_k has degree $m = k(p-1)$ on every line of $\mathcal{M}_{\mathrm{red}}^{(n)}$.

Proof (not formalized). Parts (a) and (b) are [6, Lemma 6.6] with 3, 11 and $\Gamma^+(\lambda_0^{k+1})$ replaced by p, ℓ and $\Gamma^+(\lambda_0^j)$, and with L_k in place of L_2; its proof uses the uniformization theorem of Rapoport and Zink [33, Theorem 6.30], the Hasse principle for the group of the moduli problem, Tate's theorem [37], the type number one of I, the norm theorem of Hasse and Schilling [40, Theorem III.4.1] and strong approximation, none of which depends on p or on the weight. It also needs a supersingular point on S, which exists because $h^0(Y, L_k|_Y) \geq (k-1)^2 c_j > 0$ (Lemma 6.3), so $Y \neq \emptyset$. There the identification is chosen by the Skolem–Noether theorem: the action of B on U_0, composed with the main involution, becomes the isomorphism $B \otimes_E \mathbb{Q}_{p^2} \cong M_2(\mathbb{Q}_{p^2})$ of Section 6.1, composed with σ when the embedding of $\mathcal{O}_E \otimes \mathbb{Z}_p$ into W that defines the grading differs from the identification of the completion of E at p with $\mathbb{Q}_{p^2}$ used there [6, proof of Lemma 6.6(a)]. In that case we compose the identification with σ on coordinates. This makes Ψ σ-semilinear and turns the action of Γ^+ on $\mathbb{Q}_{p^2}^2$,

and so on $\mathcal{T}$, into the standard one. The element -1 acts trivially on $\mathcal{M}^{(n)}_{\mathrm{red}}$ and on $L_k = (\omega_1 \otimes \omega_2)^{\otimes k}$, by $(-1)^{2k} = 1$, so $I[1/p]^1$ acts through Γ^+.

(c) Since $\Gamma^+(\lambda_0^j)$ acts freely, every line maps isomorphically onto a component of Y, compatibly with L_k. Let C be a component of Z_1, and $a = \deg \omega_1|_C$, $b = \deg \omega_2|_C$. The section h_2 vanishes on C exactly at its p^2+1 superspecial points, simply, because the crossings are ordinary double points of the reduced curve $Z_1 + Z_2$; so $pa - b = p^2 + 1$. The components of Z_1 are disjoint and $Z_1 \sim p\omega_2 - \omega_1$ by (G3), so $C^2 = Z_1 \cdot C = -a + pb$; and adjunction with $K_S = 2H - D$ and $D \cdot C = 0$ gives $-2 = (K_S + C) \cdot C = 2a + 2b - a + pb = a + (p+2)b$. Hence $(a, b) = (p, -1)$, and $\deg L_k|_C = k(a+b) = k(p-1)$. The components of Z_2 are symmetric. □

A *geometric datum* on $\mathcal{T}$ consists of a smooth rational curve C_v over $\bar{\mathbb{F}}_p$ and a line bundle L_v of degree m on it for every vertex v, a point $P_{v,e} \in C_v$ for every edge e at v (its *special points*), and for every edge $e = \{v, w\}$ an isomorphism φ_e between the fibres of L_v at $P_{v,e}$ and of L_w at $P_{w,e}$. For a group Γ acting on the datum and freely on $\mathcal{T}$, its *Tanner code* on $\Gamma\backslash\mathcal{T}$ is the space of Γ-invariant families of sections $s_v \in H^0(C_v, L_v)$ whose values at the two special points of each edge correspond under φ_e. By Lemma 6.4, the lines of $\mathcal{M}^{(n)}_{\mathrm{red}}$ with the restrictions of L_k, their superspecial points and the identifications of the fibres of L_k at the crossings form a geometric datum D_{geo} on $\mathcal{T}$, the same for every j.

Lemma 6.5 (Sections are Tanner codes)**.** Not formalized; a complete proof is given below.

For every subgroup $\Gamma \subseteq \Gamma^+(\lambda_0)$ of finite index, $H^0(\Gamma\backslash\mathcal{M}^{(n)}_{\mathrm{red}}, L_k)$ is isomorphic to the Tanner code of D_{geo} on $\Gamma\backslash\mathcal{T}$.

Proof (not formalized). This is [6, Lemma 6.7]: the curve $\Gamma\backslash\mathcal{M}^{(n)}_{\mathrm{red}}$ has the lines as components and ordinary double points exactly at the superspecial points, so by the normalization sequence a section of L_k is a family of sections on the components whose values at each node correspond. □

6.4 Rigidity and the twisted model over $\mathbb{F}_{p^2}$

Let $G = \mathrm{SL}_2(\mathbb{Q}_{p^2})$. It acts on $\mathcal{T}$ without inversions, transitively on the vertices of each parity and on the edges. Let $v_0 = [\mathbb{Z}_{p^2}^2]$, $v_1 = [\mathbb{Z}_{p^2}e_1 \oplus p\mathbb{Z}_{p^2}e_2]$ and $e_0 = \{v_0, v_1\}$. The stabilizers are $K_0 = \mathrm{SL}_2(\mathbb{Z}_{p^2})$, $K_1 = g_1 K_0 g_1^{-1}$ with $g_1 = \mathrm{diag}(1, p)$, and the Iwahori subgroup $K_0 \cap K_1$ of matrices with lower left entry divisible by p. Let $\rho_0 \colon K_0 \to \mathrm{SL}_2(\mathbb{F}_{p^2})$ be reduction modulo p and $\rho_1(g) = \rho_0(g_1^{-1} g g_1)$ on K_1; then ρ_0 maps the Iwahori subgroup onto the upper and ρ_1 onto the lower triangular matrices, with the same diagonal: if $g = \left(\begin{smallmatrix} a & b \\ pc & d \end{smallmatrix}\right)$, then $\rho_0(g) = \left(\begin{smallmatrix} \bar{a} & \bar{b} \\ 0 & \bar{d} \end{smallmatrix}\right)$ and $\rho_1(g) = \left(\begin{smallmatrix} \bar{a} & 0 \\ \bar{c} & \bar{d} \end{smallmatrix}\right)$, with $\bar{a}\bar{d} = 1$. Let σ be the Frobenius of $\mathbb{F}_{p^2}$, applied to matrices entrywise.

A *G-equivariant* geometric datum carries an action of G compatible with all its pieces, such that the stabilizer K of each vertex v acts on (C_v, L_v) through the reduction of K modulo p, that is, through $\mathrm{SL}(\Lambda_v/p\Lambda_v) \cong \mathrm{SL}_2(\mathbb{F}_{p^2})$ for a lattice Λ_v of v. Its *type* at v_0 (respectively v_1) is *standard* if C_v with this action is isomorphic to $\mathbb{P}^1_{\bar{\mathbb{F}}_p}$ with the action through ρ_0 (respectively ρ_1), and *twisted* if it is isomorphic to $\mathbb{P}^1_{\bar{\mathbb{F}}_p}$ with the action through $\sigma \circ \rho_0$ (respectively $\sigma \circ \rho_1$).

The twisted model. Let $m = k(p-1)$. At v_0 take $C_0 = \mathbb{P}^1_{\mathbb{F}_{p^2}}$ with K_0 acting through ρ_0, at v_1 take $C_1 = \mathbb{P}^1_{\mathbb{F}_{p^2}}$ with K_1 acting through $\sigma \circ \rho_1$, and on both the bundle $\mathcal{O}(m)$, whose sections are the binary forms of degree m, with its natural linearization. The special points of e_0 are the points $[e_1] \in C_0$ and $[e_2] \in C_1$: they are the unique fixed points of the images of the Iwahori subgroup, upper and lower triangular. The fibre of $\mathcal{O}(m)$ at $[e_1]$ has the basis $(e_1^\vee)^{\otimes m}$, on which the Iwahori

subgroup acts by $\bar{a}^{-m}$; the fibre at $[e_2]$ has the basis $(e_2^\vee)^{\otimes m}$, on which it acts by $\sigma(\bar{d})^{-m} = \bar{a}^{pm}$. Since $\bar{a}^{(p+1)m} = \bar{a}^{k(p^2-1)} = 1$, the identification $\varphi_{e_0}\colon (e_1^\vee)^{\otimes m} \mapsto (e_2^\vee)^{\otimes m}$ is equivariant. Transporting these data by G, which is possible because G acts on $\mathcal{T}$ with the edge e_0 as a fundamental domain and every piece is equivariant for the relevant stabilizer, defines a G-equivariant geometric datum D_{tw} of degree m, standard at the vertices of the parity of v_0 and twisted at the others. Every piece of it is defined over $\mathbb{F}_{p^2}$, and G acts through $\mathbb{F}_{p^2}$-rational maps. The element -1 acts trivially, since m is even, so subgroups of $\mathrm{PSL}_2(\mathbb{Q}_{p^2})$ act on it. Let D_{tw}^σ be its image under σ, twisted at the vertices of the parity of v_0 and standard at the others.

Restrict D_{tw} to $\Gamma_B = \Gamma^+(\lambda_0)$, which acts freely on edges. Choose an $\mathbb{F}_{p^2}$-basis of the fibre at one edge of each Γ_B-orbit and transport it by Γ_B. Every fibre becomes $\mathbb{F}_{p^2}$, and evaluation of sections at the special points turns the datum into a Γ_B-equivariant local datum over $\mathbb{F}_{p^2}$ in the sense of Section 2.3. Its local codes are the codes of Lemma 2.5 with $q = p^2$: evaluation is injective because $m \le p^2 - 1$, so the Tanner code of the geometric datum on $\Gamma\backslash\mathcal{T}$ is the Tanner code $T_{\mathrm{tw}}(\Gamma\backslash\mathcal{T})$ of Section 2.3 for every $\Gamma \subseteq \Gamma_B$ of finite index, and the local checks interpolate on $t = p^2 - m$ coordinates. Extending scalars to $\bar{\mathbb{F}}_p$ gives the Tanner code of $D_{\mathrm{tw}} \otimes \bar{\mathbb{F}}_p$, which is therefore $T_{\mathrm{tw}}(\Gamma\backslash\mathcal{T}) \otimes_{\mathbb{F}_{p^2}} \bar{\mathbb{F}}_p$.

Lemma 6.6 (The geometric datum is twisted)**.** Not formalized; a complete proof is given below.

The datum D_{geo} extends to a G-equivariant geometric datum of degree $m = k(p-1)$, under the identification Ψ of U_0 with $\mathbb{Q}_{p^2}^2$ of Lemma 6.4, whose action extends that of Γ^+; its type is standard at the vertices of one parity and twisted at the others.

Proof (not formalized). The group J consists of the automorphisms of the isocrystal that commute with F, respect the $\mathcal{O}_E$-grading and multiply the polarization by a factor in $\mathbb{Q}_p^\times$ [24, (4.5)]. Such an automorphism is determined by its action on U_0, because it commutes with Φ, and any $g \in \mathrm{GL}(U_0)$ with $\det g \in \mathbb{Q}_p^\times$ extends to one. So conjugation by Ψ identifies J with $\{g \in \mathrm{GL}_2(\mathbb{Q}_{p^2}) : \det g \in \mathbb{Q}_p^\times\}$, as in [24, (5.12)]; if Ψ is σ-semilinear, conjugation by it applies σ to determinants, which preserves $\mathbb{Q}_p^\times$. In particular $G \subseteq J^0$. The group J acts on $\mathcal{M}_{\mathrm{red}}^{(n)}$ and on L_k compatibly with the crossings and with the identification of the dual graph with $\mathcal{T}$ [6, Section 6.3], so D_{geo} is G-equivariant, and the action of Γ^+ of Lemma 6.4 is the restriction.

Let Λ be a lattice of the first kind and K its stabilizer in G. A point of C_Λ is a Dieudonné lattice $M = M_0 \oplus M_1$ with $M_0 = \Lambda \otimes W$, for which the index 0 is critical, $F^2 M_0 = pM_0$, equivalently $V^2 M_0 = pM_0$ [24, Definition 4.1, Lemma 4.3 and (4.9)]. The group K acts on $C_\Lambda = \mathbb{P}^1(\Lambda/p\Lambda)$ through its action on $\Lambda/p\Lambda$ [24, Lemma 4.3]. After a change of coordinates this is the action through ρ_i if Ψ is linear, and through $\sigma \circ \rho_i$ if Ψ is σ-semilinear, where v_i is the vertex among v_0, v_1 of the parity of $[\Psi\Lambda]$; in the second case Ψ induces a σ-semilinear bijection of $\Lambda/p\Lambda$ onto $\Psi\Lambda/p\Psi\Lambda$. If $g \in K$ is congruent to 1 modulo $p\Lambda$, then g fixes C_Λ pointwise. It also acts trivially on VM/pM, the fibre of the Hodge bundle at the point M in the contravariant Dieudonné theory of [24]. Indeed $(g-1)M_0 \subseteq pM_0$, and, since g commutes with F and V and $FM_1 \subseteq M_0$ [24, (4.6)], $(g-1)M_1 = F^{-1}(g-1)FM_1 \subseteq F^{-1}pM_0 = VM_0$. Hence $(g-1)VM_1 = V(g-1)M_1 \subseteq V^2 M_0 = pM_0$ and $(g-1)VM_0 = V(g-1)M_0 \subseteq pVM_0 \subseteq pM_1$. So g acts trivially on the fibres of ω_1, ω_2 and L_k over C_Λ, and the stabilizer acts on (C_Λ, L_k) through its reduction modulo p. For lines of the second kind the same holds with the roles of M_0 and M_1 exchanged, and the action on $C_{\Lambda'}$, in the coordinates of U_0 transported by Φ, is conjugated by σ because Φ is σ-semilinear and commutes with J [6, Lemma 6.9, steps (1)–(2)]. So if Ψ is linear, the type is standard at the vertices that carry lines of the first kind and twisted at the others; if Ψ is σ-semilinear, both are replaced by their images under σ, as in [6, Lemma 6.9, step (5)]. Lines of the first kind sit at the vertices of one parity and lines of the second kind at the others, so in both cases the type is standard at one parity and twisted at the other. By Lemma 6.4(c) the degree is m. □

Lemma 6.7 (Rigidity)**.** Not formalized; a complete proof is given below.

Two G-equivariant geometric data of degree m with the same types at v_0 and at v_1 are isomorphic, G-equivariantly. If moreover $-1 \in G$ acts trivially on both data, then for every subgroup $\Gamma \subseteq \mathrm{PSL}_2(\mathbb{Q}_{p^2})$ acting freely on $\mathcal{T}$ with finite quotient, their Tanner codes on $\Gamma\backslash\mathcal{T}$ are isomorphic.

Proof (not formalized). Let D and D' be the data. (1) At v_i the types agree, so there is a K_i-equivariant isomorphism $\psi_i \colon C_{v_i} \to C'_{v_i}$. (2) The bundles L_{v_i} and $\psi_i^* L'_{v_i}$ have the same degree, hence are isomorphic; the two actions of K_i on L_{v_i} that result factor through $\mathrm{SL}_2(\mathbb{F}_{p^2})$ and cover the same action on the curve, so they differ by a character of $\mathrm{SL}_2(\mathbb{F}_{p^2})$ with values in $\bar{\mathbb{F}}_p^\times$. Such a character is trivial: $\mathrm{SL}_2(\mathbb{F}_{p^2})$ is generated by the elementary matrices $\left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ a & 1 \end{smallmatrix}\right)$, which have order dividing p, and $\bar{\mathbb{F}}_p^\times$ has no element of order p. So the isomorphism of bundles can be chosen K_i-equivariant. (3) The special point P_{v_0,e_0} is fixed by the Iwahori subgroup, whose image in the action on C_{v_0} is a Borel subgroup of $\mathrm{SL}_2(\mathbb{F}_{p^2})$; it contains a nontrivial unipotent element, which has exactly one fixed point on $\mathbb{P}^1$. So $\psi_0(P_{v_0,e_0}) = P'_{v_0,e_0}$, and likewise at v_1. (4) The two identifications of fibres at e_0 now differ by a scalar; rescaling the isomorphism of bundles at v_1 makes them agree. (5) Transport the isomorphisms by G, setting $\psi_{gv_i} = g\psi_i g^{-1}$; this is well defined because ψ_i is K_i-equivariant. Every edge is ge_0 for some $g \in G$, so the special points and identifications match at every edge, and the transported maps form a G-equivariant isomorphism of data. It descends to $\Gamma\backslash\mathcal{T}$ and maps one Tanner code onto the other. □

Remark 6.8 (The twist is forced)**.** For $k \neq (p+1)/2$, equivariance alone forces the twist, independently of the computation with lines of the second kind in the proof of Lemma 6.6: a G-equivariant geometric datum of degree m whose types at v_0 and v_1 are each standard or twisted is standard at one of them and twisted at the other. In a datum that is standard at both v_0 and v_1, the Iwahori subgroup acts on the fibres at the two special points of e_0 by $\bar{a}^{-m}$ and $\bar{d}^{-m} = \bar{a}^m$, and an equivariant identification exists only if $\bar{a}^{2m} = 1$ for all $\bar{a} \in \mathbb{F}_{p^2}^\times$, that is, if $p+1$ divides $2k$. For $2 \le k \le p-2$ this means $k = (p+1)/2$. The same holds when both types are twisted. The weight $k = (p+1)/2$ of the twelfth paper, where $p = 3$ and $k = 2$, is exactly the exceptional case, which is why that paper needed an explicit identification of the type [6, Lemma 6.9].

Theorem 6.9 (Theorem G)**.** Not formalized; a complete proof is given below.

Let $p \ge 7$ be a prime inert in $E = \mathbb{Q}(\sqrt{5})$, let $2 \le k \le p-2$, $m = k(p-1)$ and $t = p^2 - m$, and let λ_0 be a prime of $\mathcal{O}_E$ of prime norm $\ell \ge 7$. The twisted datum D_{tw} is a $\Gamma^+(\lambda_0)$-equivariant local datum over $\mathbb{F}_{p^2}$ on the (p^2+1)-regular tree, whose local codes are generalized Reed–Solomon codes of length p^2+1 and dimension $m+1$, with local checks interpolating on t coordinates. For every $j \ge 1$,

$$\dim_{\mathbb{F}_{p^2}} T_{\mathrm{tw}}(X_j) \;\ge\; (k-1)^2 c_j = \frac{(k-1)^2}{p^2+1}\,|X_j^1|.$$

Proof (not formalized). The first sentence summarizes the construction of the twisted model. By Lemma 6.6, D_{geo} is a G-equivariant datum of degree m, standard at the vertices of one parity and twisted at the others. If the vertices of the parity of v_0 are standard, Lemma 6.7 gives $D_{\mathrm{geo}} \cong D_{\mathrm{tw}} \otimes \bar{\mathbb{F}}_p$; otherwise $D_{\mathrm{geo}} \cong D_{\mathrm{tw}}^\sigma \otimes \bar{\mathbb{F}}_p$, whose Tanner codes are the images under σ of those of D_{tw}. In both cases, by Lemma 6.4(b), Lemmas 6.3 and 6.5,

$$\dim_{\mathbb{F}_{p^2}} T_{\mathrm{tw}}(X_j) = \dim_{\bar{\mathbb{F}}_p} T_{D_{\mathrm{geo}}}(X_j) = h^0(Y, L_k|_Y) \ge (k-1)^2 c_j.$$

Finally $|X_j^1| = (p^2+1)c_j$ by Lemma 6.1(b). □

Counting constraints gives only $\dim T_{\mathrm{tw}}(X_j) \geq (p^2+1-2t)c_j$, since each vertex imposes t conditions. For t near $(2-\sqrt{3})(p^2+1)$ this is less than half the length, while Theorem G gives a rate of at least about $(\sqrt{3}-1)^2 \approx 0.536$. The excess comes from the geometry of the Goren–Oort configuration and is a phenomenon of characteristic p (Section 10).

Remark 6.10 (A numerical check)**.** The script `tools/twisted_tanner.py` builds the twisted datum for $p = 7$ and a prime λ_0 above $\ell = 11$, where $c_1 = 11$, and computes $\dim_{\mathbb{F}_{49}} T_{\mathrm{tw}}(X_1)$ exactly. For $k = 2, 3, 4, 5$ it finds 22, 44, 111 and 176, against the bounds 11, 44, 99 and 176 of Theorem G. As controls, the untwisted gluing, which is equivariant only for $k = (p+1)/2 = 4$ (Remark 6.8), gives $60 < 99$ there, and random gauges on the edges give the counting bound $\max(0, (p^2+1-2t)c_1)$ or one dimension more. Here $\ell < 4p$, so X_1 is not simple, which Theorem G does not need. The computation is not used in any proof.

7 Question 9.1 has a negative answer

7.1 The explicit instance

Take $p = 127$, which is inert in E because $127 \equiv 2 \pmod 5$; the weight $k = 91$, so $m = 91 \cdot 126 = 11466$ and $t = 127^2 - m = 4663$; and a prime λ_0 of $\mathcal{O}_E$ above $\ell = 509$, which splits in E because $509 \equiv 4 \pmod 5$. Then $\ell > 4p = 508$, and

$$|\mathrm{PSL}_2(\mathbb{F}_{509})| = 65\,935\,860, \qquad c_1 = 1\,098\,931, \qquad n = 2c_1 = 2\,197\,862.$$

Put $R = 2n - 1$ and

$$\Delta = \beta_{p^2}(R) - 1 = \frac{2(p^{2(R+1)} - 1)}{p^2 - 1} - 2 < p^{4n} = 127^{8\,791\,448}.$$

Theorem 7.1 (Theorem A)**.** Lean (partial):
`AboveHalf.not_Q91At_of_tower` [`AboveHalfRate/Main.lean`]
Not formalized: the arithmetic input (Theorem 6.9), the probabilistic step of Theorem 3.2, the passage from a lift selection to separated checks (Propositions 4.3 and 4.4), the ball count (Lemma 2.2) and the girth bound (Lemma 6.2), the icosian group and its congruence subgroups (Lemma 6.1) and the Reed–Solomon local codes (Lemma 2.5).

With these parameters and $X_j = \Gamma^+(\lambda_0^j)\backslash\mathcal{T}$ $(j \geq 1)$, there are recovery relations N_j on the edge sets X_j^1 such that:

(a) every incidence degree is at most Δ;

(b) every coordinate has paired distance at least $(j \log 509 - \log 4)/(n \log 127)$, which tends to infinity;

(c) the $\mathbb{F}_{127^2}$-linear Tanner code $T_{\mathrm{tw}}(X_j)$ is a storage code for N_j of rate at least $8100/16130 > 0.5021$.

Hence Question 2.1 has a negative answer at the degree bound Δ, with the fixed alphabet $\mathbb{F}_{127^2}$ and linear codes.

Outline of the formal proof. Let $\Gamma = \Gamma^+(\lambda_0)$. By Lemmas 6.1 and 6.2, Γ acts freely and without inversions with $s(\Gamma) \geq 4$, and $X_1 = \Gamma\backslash\mathcal{T}$ is a simple 16130-regular graph on n vertices. By Corollary 3.7 it has a 4640-degenerate lift selection, so by Proposition 4.4 the tree carries a twisted

separated structure for Γ of degeneracy 4640 and depth R. By Theorem 6.9 the twisted datum is Γ-equivariant over $\mathbb{F}_{127^2}$, its local checks interpolate on $t = 4663 \geq 4640$ coordinates, and $\dim T_{\mathrm{tw}}(X_j) \geq 8100\, c_j = \frac{8100}{16130}|X_j^1|$. The balls of radius R have $\beta_{p^2}(R)$ edges (Lemma 2.2). The subgroups $\Gamma_j = \Gamma^+(\lambda_0^j)$ lie in Γ, have finite quotients X_j with $|X_j^1| = 16130\, c_j \to \infty$, and displacement $s(\Gamma_j) \geq 2(j \log 509 - \log 4)/\log 127 \to \infty$ (Lemma 6.2). Theorem T(a) and (b) (Theorem 5.3) give the claims, with paired distance at least $s(\Gamma_j)/R$, which is at least the bound in (b). The rate exceeds $1/2$ because $2 \cdot 8100 > 16130$. The Lean theorem `AboveHalf.not_Q91At_of_tower` is the last step, from the checks and the tower to the negation of `MergeDepth.Q91At`(Δ), and `AboveHalf.Numbers.params` checks the arithmetic of the parameters, including $8100/16130 > 0.5021$ and $4n = 8\,791\,448$. □

The incidence degree is astronomically large because the private checks of Lemma 4.1 may reach the whole down-set, whose depth we bound only by the number $2n$ of ranks. The rate margin is small, 0.00217 above one half, and so is the margin in Corollary 3.7: 23 out of $t = 4663$. Larger primes give rates up to the limit of the method.

Remark 7.2 (Numerical: the least prime)**.** Numerical, not used in any proof. Among the primes p inert in $\mathbb{Q}(\sqrt{5})$, $p = 127$ is the least for which the method fits as we apply it: with ℓ the least prime $\equiv \pm 1 \pmod 5$ above $4p$, the optimal λ in (2) and the least weight k with $2(k-1)^2 > p^2 + 1$, the redundancy $p^2 - k(p-1)$ reaches $\lceil \tau(p^2+1) + \lambda \rceil$ for no inert $p < 127$. Smaller primes are not excluded for other selections or other bases.

7.2 All large inert primes

Theorem 7.3 (Theorem A*)**.** Not formalized; a complete proof is given below.
For every $\varepsilon > 0$ there is p_0 such that for every prime $p \geq p_0$ inert in $\mathbb{Q}(\sqrt{5})$ there are a degree bound Δ_p, a sequence of recovery relations with incidence degree at most Δ_p on which every paired distance tends to infinity, and $\mathbb{F}_{p^2}$-linear storage codes on them of rate at least $(\sqrt{3}-1)^2 - \varepsilon = 4 - 2\sqrt{3} - \varepsilon \approx 0.536 - \varepsilon$.

Proof (not formalized). Let $\tau = 2 - \sqrt{3}$, so $1 - \tau = \sqrt{3} - 1$, and choose η with $0 < \eta < 1 - \tau$ and $(1 - \tau - \eta)^2 > 4 - 2\sqrt{3} - \varepsilon/2$. For p large choose a prime $\ell \equiv \pm 1 \pmod 5$ with $4p < \ell \leq 8p$; it exists for all large p by the prime number theorem for arithmetic progressions. Let λ_0 be a prime of $\mathcal{O}_E$ above ℓ. Then X_1 is simple (Lemma 6.2) and has $n = \ell(\ell^2-1)/60 \leq 9p^3$ vertices. Let $k = \lfloor (1 - \tau - \eta)p \rfloor$, so $2 \leq k \leq p - 2$ for large p, and

$$t = p^2 - k(p-1) \geq p^2 - (1 - \tau - \eta)p^2 = (\tau + \eta)p^2.$$

Let $\lambda_p = 1 + \lambda_*$, where λ_* is the positive root of $\lambda^2 = 2\log(2n)(\tau(1-\tau)(p^2+1) + \lambda/3)$; then (2) holds and $\lambda_p = O(p\sqrt{\log p})$. Theorem 3.2, applied to X_1, which is regular of degree $p^2 + 1$, gives a lift selection of degeneracy at most $\tau(p^2+1) + \lambda_p + 1 \leq t$ for large p. As in the proof of Theorem 7.1, Theorem T and Theorem G give storage codes of rate at least

$$\frac{(k-1)^2}{p^2+1} \geq \frac{\big((1-\tau-\eta)p - 2\big)^2}{p^2+1} \longrightarrow (1-\tau-\eta)^2 > 4 - 2\sqrt{3} - \varepsilon/2.$$

□

The binary blow-up of Section 8 gives binary codes with the same limit of rates, because the factor $\log_2(p^2)/\lceil \log_2(p^2) \rceil$ tends to 1. The limit $4 - 2\sqrt{3}$ is that of the method, not of the question: it comes from the constant of Theorem 3.2 and the tensor shape $(1 - t/(p^2+1))^2$ of the lower bound on the rate in Theorem 6.9; see Section 10.

8 Binary codes

The fifth paper singles out the fixed binary alphabet. A code over a larger alphabet becomes binary by writing every symbol in bits, and the recovery relation is enlarged accordingly; this is the blow-up of guessing-number theory [13], and we only record the bookkeeping of degrees and paired distances.

Lemma 8.1 (Blow-up)**.** Lean: `AboveHalf.blowUp`, `AboveHalf.blowCode`, `AboveHalf.isStorageCode_blowCode`, `AboveHalf.card_blowCode`, `AboveHalf.degree_blowUp_le`, `AboveHalf.edist_le_blowUp`, `AboveHalf.shortPairedCoordinates_blowUp_subset` [`AboveHalfRate/BlowUp.lean`]

Let N be a recovery relation on V, $\mathcal{C} \subseteq \mathcal{A}^V$ a storage code for N, and $\iota\colon \mathcal{A} \to \{0,1\}^b$ injective. On $V \times [b]$ let $N^{(b)}(p,i) = N(p) \times [b]$, and let $\mathcal{C}^{(b)}$ be the set of words $(p,i) \mapsto \iota(x_p)_i$ for $x \in \mathcal{C}$. Then $\mathcal{C}^{(b)}$ is a binary storage code for $N^{(b)}$ with $|\mathcal{C}^{(b)}| = |\mathcal{C}|$; every incidence degree of $N^{(b)}$ is at most b times the largest incidence degree of N; and the paired distance of (p,i) in $N^{(b)}$ is at least that of p in N.

Outline of the formal proof. Words of $\mathcal{C}^{(b)}$ that agree on $N(p) \times [b]$ come from words of $\mathcal{C}$ that agree on $N(p)$, because ι is injective, so they agree at p and at every bit of p. Encoding is injective on words. The recovery set of (p,i) has $b|N(p)|$ elements, and (q,j) lies in b times as many recovery sets as q. Forgetting the bit index is a homomorphism from the paired incidence graph of $N^{(b)}$ to that of N, and homomorphisms do not increase distances. □

The rate of $\mathcal{C}^{(b)}$ is $\log_2|\mathcal{C}|/(b|V|) = (\log_2|\mathcal{A}|/b)$ times the rate of $\mathcal{C}$. For $\mathcal{A} = \mathbb{F}_{127^2}$ we take $b = 14$, since $127^2 = 16129 \le 2^{14}$; the loss is the factor $\log_2(16129)/14 > 0.9983$.

Corollary 8.2 (Corollary A′)**.** Lean (partial): `AboveHalf.not_Q91BinaryAt_of_tower` [`AboveHalfRate/Main.lean`]

Not formalized: the inputs of Theorem 7.1 listed there.

The blow-up with $b = 14$ of the relations and codes of Theorem 7.1 gives binary storage codes of rate at least

$$\frac{8100}{16130} \cdot \frac{\log_2 16129}{14} > 0.5013$$

on recovery relations of incidence degree at most 14Δ on which every paired distance tends to infinity. Hence Question 2.1 has a negative answer for the fixed binary alphabet, at the degree bound 14Δ.

Outline of the formal proof. Apply Lemma 8.1 to the relations and codes of Theorem 7.1 with an injection $\mathbb{F}_{127^2} \to \{0,1\}^{14}$; this is Theorem T(c). The Lean theorem `AboveHalf.not_Q91BinaryAt_of_tower` is the last step, and `AboveHalf.Numbers.binary_rate` checks that $127^2 \le 2^{14}$ and $\frac{8100}{16130} \cdot \frac{\log_2 16129}{14} > 0.5013$, using $\log_2 16129 > 615/44$, that is, $2^{615} < 16129^{44}$. □

The binary codes are not linear over $\mathbb{F}_2$: they are $\mathbb{F}_{127^2}$-linear codes written in bits. Whether Question 2.1 has a negative answer for $\mathbb{F}_2$-linear codes is open (Section 11).

9 The undirected crux

The fourth paper of the series asks the undirected form of Question 2.1 [5, Question 6.1]. A storage code on a simple graph H is a storage code for its neighbourhood relation, which assigns to each vertex the set of its neighbours [5, Section 2.3]. A sequence of graphs is *locally bipartite* if, for each fixed ℓ, only a vanishing fraction of the vertices lie on odd cycles of length at most ℓ.

Question ([5, Question 6.1]). *Let (H_n) be locally bipartite graphs of maximum degree at most Δ. Does every storage code on H_n have rate at most $\frac{1}{2} + o(1)$, over every alphabet?*

Theorem T answers it as well, once the directions of the recovery relations are forgotten. The *symmetrization* of a relation N on V is the simple graph $H(N)$ on V in which two different coordinates p, q are adjacent when $p \in N(q)$ or $q \in N(p)$. If N has no loops, that is, $q \notin N(q)$ for every q, then every recovery set $N(q)$ lies in the neighbourhood of q in $H(N)$, so every storage code for N is a storage code on $H(N)$. The degree of q in $H(N)$ is at most the sum of its two incidence degrees.

Proposition 9.1 (Symmetrized quotients). Lean: `AboveHalf.CheckSystem.proj_not_mem_qrel`, `AboveHalf.CheckSystem.symGraph`, `AboveHalf.CheckSystem.degree_symGraph_le`, `AboveHalf.CheckSystem.le_mul_of_odd_walk`, `AboveHalf.CheckSystem.isStorageCode_nbhd_symGraph` [`AboveHalfRate/Descent.lean`]

In the setting of Proposition 5.1, suppose that every element of $\Gamma' \setminus \{1\}$ moves every edge by at least $s > R$. Then the descended relation N' has no loops, and its symmetrization $H' = H(N')$ satisfies:

(a) every vertex has degree at most $2(\beta - 1)$;

(b) every closed walk of odd length k in H' satisfies $kR \geq s$;

(c) every storage code for N' is a storage code on H'.

Outline of the formal proof. (a) follows from Proposition 5.1(b). (b) An edge of H' joins πe to πf for a check h_e and an edge $f \neq e$ of its support, in one direction or the other. By (b) and (c) of Proposition 4.3, e and f have different classes and $\operatorname{dist}(e, f) \leq R$. As in the proof of Proposition 5.1(c), a closed walk of length k at πa therefore lifts to a walk from a to some γa with $\gamma \in \Gamma'$, along which the class changes at every step and the edge moves by at most kR. For odd k this gives $\operatorname{cls}(\gamma a) = \operatorname{cls}(a) + 1$, so $\chi(\gamma) = 1$, hence $\gamma \neq 1$ and $s \leq \operatorname{dist}(a, \gamma a) \leq kR$. (c) A loop $\pi e \in N'(\pi e)$ would give an edge $f = \gamma e \neq e$ in the support of h_e; then $\gamma \neq 1$ and $s \leq \operatorname{dist}(e, \gamma e) \leq R$, which is excluded. So N' has no loops, and (c) follows. □

For binary codes, the blow-up of Section 8 has to keep odd cycles long, so the copies of a vertex are not joined to each other.

Lemma 9.2 (Blow-up of a graph). Lean: `AboveHalf.graphBlowUp`, `AboveHalf.nbhd_graphBlowUp`, `AboveHalf.isStorageCode_graphBlowUp`, `AboveHalf.degree_graphBlowUp_le`, `AboveHalf.graphBlowUpHom`, `AboveHalf.exists_walk_of_graphBlowUp` [`AboveHalfRate/Undirected.lean`]

Let H be a simple graph on V, $\mathcal{C} \subseteq \mathcal{A}^V$ a storage code on H, and $\iota \colon \mathcal{A} \to \{0,1\}^b$ injective. Let $H^{(b)}$ be the graph on $V \times [b]$ in which (p, i) and (q, j) are adjacent when p and q are. Then the binary code $\mathcal{C}^{(b)}$ of Lemma 8.1 is a storage code on $H^{(b)}$; every degree of $H^{(b)}$ is at most b times the largest degree of H; and every walk in $H^{(b)}$ projects to a walk in H of the same length.

Outline of the formal proof. The neighbourhood of (p, i) in $H^{(b)}$ is $N_H(p) \times [b]$, which is the recovery set of (p, i) in the blow-up of the neighbourhood relation of H, so Lemma 8.1 applies. Forgetting the second coordinate is a homomorphism $H^{(b)} \to H$, and homomorphisms map walks to walks of the same length. □

Corollary 9.3 (Corollary U). Lean (partial): `AboveHalf.undirected_tower`, `AboveHalf.undirected_tower_binary` [`AboveHalfRate/Undirected.lean`]

Not formalized: the inputs of Theorem 7.1 listed there.

Keep the notation of Theorem 7.1, and let H_j be the symmetrization of N_j, for the levels j with $s(\Gamma^+(\lambda_0^j)) > R$, which include all large j.

(a) *The graph H_j has maximum degree at most 2Δ, and every closed walk in H_j of odd length k has $k \geq (j \log 509 - \log 4)/(n \log 127)$. So for every K, eventually no vertex of H_j lies on an odd cycle of length at most K.*

(b) *The $\mathbb{F}_{127^2}$-linear code $T_{\mathrm{tw}}(X_j)$ is a storage code on H_j of rate at least $8100/16130 > 0.5021$.*

(c) *The blow-ups $H_j^{(14)}$ have maximum degree at most 28Δ, satisfy the same bound on closed walks of odd length, and carry binary storage codes of rate greater than 0.5013.*

Hence the fourth paper's Question 6.1 has a negative answer, at the degree bound 2Δ over the alphabet $\mathbb{F}_{127^2}$ and at the degree bound 28Δ over the binary alphabet.

Outline of the formal proof. For the levels with $s(\Gamma^+(\lambda_0^j)) > R$, apply Proposition 9.1 with $\Gamma' = \Gamma^+(\lambda_0^j)$ and $\beta - 1 = \Delta$. The bound on odd closed walks is $s(\Gamma^+(\lambda_0^j))/R$, which is at least the bound in (a), as in Theorem 7.1(b), and it tends to infinity, since $s(\Gamma^+(\lambda_0^j)) \geq 2(j \log 509 - \log 4)/\log 127$ by Lemma 6.2. Part (b) is Theorem 7.1(c) with Proposition 9.1(c). For (c), apply Lemma 9.2 with an injection $\mathbb{F}_{127^2} \to \{0,1\}^{14}$; the rate is computed as in Corollary 8.2. The Lean theorems `AboveHalf.undirected_tower` and `AboveHalf.undirected_tower_binary` prove, from the hypotheses of `AboveHalf.not_Q91At_of_tower` other than $\rho > 1/2$, and for the binary one an injective encoding, that for every K, eventually the graphs have the stated degrees, no closed walk of odd length at most K, and storage codes of rate at least ρ, respectively $\rho \log_2 |\mathbb{F}|/b$. □

By Theorem 7.3, the same construction gives, for every large prime p inert in $\mathbb{Q}(\sqrt{5})$, locally bipartite graphs of bounded degree with $\mathbb{F}_{p^2}$-linear storage codes of rate at least $4 - 2\sqrt{3} - \varepsilon$ for $p \geq p_0(\varepsilon)$. Corollary 9.3 concerns arbitrary graphs. The crux of the fourth paper itself [5, Definition 4.8] is a statement about the side-information systems of oblivious machines, and Corollary 9.3 does not decide it.

10 What the counterexample changes

10.1 For lower bounds and for constructions

The fifth paper proposed Question 2.1 as a machine-free route to lower bounds: an affirmative answer, for every fixed Δ, would bound by $1/2 + o(1)$ the rate of every finite-alphabet storage code on the actual cone relations of oblivious linear-time computations [11, Proposition 9.2], and combined with cheap depth reduction this would exclude oblivious linear-time transposers. Theorems A and A′ close this route in its machine-free form: bounded degree and the paired-distance condition do not force rate $1/2$, for linear codes over a fixed field or for binary codes. Corollary U closes the undirected form of the route, the fourth paper's Question 6.1 [5], in the same way: bounded degree without short odd cycles does not force rate $1/2$ either. The conclusion of the fifth paper's Proposition 9.2 for actual cone relations is not refuted, because nothing here shows that our relations arise from tape computations. A half-rate bound for the cone relations of linear-time machines, or any bound below the rates of Theorems A and A*, must therefore use structure of tape computations beyond bounded degree and paired distances. A fixed deficit below rate 1 is not excluded by our construction (Section 11). The fifth paper records such structure, for instance its transport bound for matchings of cone relations [11, Lemmas 3.4 and 3.5]. Several later papers of the series state results conditional on an affirmative answer to Question 2.1; as implications they remain correct, but their hypothesis is now known to be false in the generality stated, and they need the native form of the hypothesis instead.

On the construction side, a linear-time transposer needs its cone relations to carry codes whose rate beats the genie cost [11, Section 2]; for genies of cost εL this means rate close to 1. The relations of Theorem A carry codes of rate above one half, which defeats the half-bound route but not this requirement; they come with an astronomically large degree, a characteristic-127 linear structure and a twist by a character of the congruence subgroup $\Gamma^{+}(\lambda_0)$, and no tape realization of them is known. Whether a linear-time oblivious machine can have cone relations with this kind of twisted, locally separated structure is the question that the counterexample leaves for the construction side.

10.2 Limits of the method

The rates guaranteed by Theorem A* approach $(\sqrt{3}-1)^2 = 4-2\sqrt{3} \approx 0.536$. Two features of the construction set this limit. Our lower bound for the Goren–Oort codes, from Riemann–Roch, is the rate $(1-t/(p^2+1))^2$ of a tensor product of two local codes, up to lower-order terms; we prove no upper bound beyond the trivial one, but a rate close to 1 would need a different source. And the separated structures of Theorem 3.2 have degeneracy at least about $(2-\sqrt{3})(p^2+1)$, since the fluid rule gives every lifted vertex expected load τ per unit degree; whether other lift selections can do better is Remark 10.1.

Remark 10.1 (Heuristic: the constant of Theorem D is optimal)**.** Heuristic, not used in any proof. The fluid rule balances the loads exactly: every lifted vertex has expected load τ per unit degree whatever its label (Lemma 3.5). For an order of the lifted vertices of a random-like graph, a lifted vertex at a given position has a definite fraction of its neighbours later, and summing the resulting constraints over all positions suggests that no lift selection of such a graph has degeneracy below $(2-\sqrt{3}-o(1))\delta$. If so, constructions of the present type, with tensor-shaped rates, cannot exceed $4-2\sqrt{3}$.

Remark 10.2 (Congruence levels are never untwisted)**.** This remark is not used elsewhere. In the setting of Theorem 7.1, no $\Gamma^{+}(\lambda_0^{j})$ is contained in the kernel of the twist χ. Otherwise the classes would descend to X_j^1, every recovery set of N_j would lie in the other class, and the signed class of the proof of Proposition 5.1(c) would separate L_x from R_x in the incidence graph; by the fifth paper's half bound for such relations [11, Lemma 4.1], every storage code would then have rate at most $1/2$, contradicting Theorem 6.9. So the arithmetic forces the twist: the subgroup $\ker\chi$ contains none of the congruence subgroups $\Gamma^{+}(\lambda_0^{j})$.

10.3 Characteristic p

Theorem G is a phenomenon of positive characteristic. Along the congruence tower the normalized dimension of the Tanner codes stays at least $(k-1)^2/(p^2+1)$, while along towers with p-group quotients the normalized dimension of the same Tanner operator converges to a rank over a division ring, as shown by Jaikin-Zapirain and Souza [22, Theorem 8.1]; the twelfth paper of the series uses this difference to show that Lück approximation fails in positive characteristic [6]. The sixth paper of the series bounds linear storage codes along towers obtained from pro-p embeddings in characteristic p, through the same theorem of Jaikin-Zapirain and Souza [10]; our tower has quotients of order a power of 509 and codes of characteristic 127, so it lies outside those results.

Remark 10.3 (Heuristic: characteristic zero)**.** Heuristic, not used in any proof. Over subfields of $\mathbb{C}$, Lück's approximation theorem [28] and its extensions [21, Section 1 and Theorem 10.1] make normalized dimensions of Tanner codes converge, along every tower of normal subgroups with trivial intersection, to the von Neumann dimension of the Tanner code on the tree, which is the same for all towers. For relations that are perfectly separated on the tree, as ours are, this common limit

appears to be at most $1/2$, by comparison with the untwisted covers, which obey the half bound. This suggests that linear codes in characteristic zero cannot replace the Goren–Oort codes, and that the half bound may hold for complex-linear codes on all bounded-degree relations with the paired-distance property; the fifth paper proves it for complex-linear codes with bounded coefficients at fixed vector width [11, Theorem 4.5].

11 Open problems

Theorem A answers Question 2.1 for linear codes over a fixed field of characteristic 127 and for binary codes, and Corollary U answers its undirected form. The following questions remain, roughly in order of their importance for linear-time transposition.

(1) *Native relations.* Does the half bound, or a fixed deficit below rate 1, hold for the actual cone relations of oblivious linear-time machines, as in the conclusion of [11, Proposition 9.2]? Which property of tape computations, beyond bounded degree and paired distances, excludes twisted separated structures, or does some machine realize them?

(2) *A deficit.* Is there, for every Δ, a $\rho(\Delta) > 0$ such that storage codes on relations of incidence degree at most Δ with the paired-distance property have rate at most $1 - \rho(\Delta) + o(1)$? The rates that our bounds guarantee approach $4 - 2\sqrt{3} \approx 0.536$; codes of rate tending to 1 on such relations would answer this negatively.

(3) *Characteristic* 2 *and* $\mathbb{F}_2$*-linear codes.* Is there a counterexample with $\mathbb{F}_2$-linear codes? The prime 2 is inert in $\mathbb{Q}(\sqrt{5})$, but the tree then has degree 5, far too small for Theorem 3.2; a construction would need trees of large degree $2^s + 1$ with a collapse of the same tensor type.

(4) *Small degree.* What is the least Δ at which Question 2.1 has a negative answer, and the least degree at which the fourth paper's Question 6.1 has one? Our bounds $\Delta < 127^{8\,791\,448}$ and 2Δ come from the depth of the order in Theorem 3.2. For recovery sets of size at most three the seventh paper's Conjecture P would give an affirmative answer to Question 2.1 [8], and for linear codes on triangle-free graphs of maximum degree three the fourth paper proves the half bound [5, Theorem 6.13].

(5) *Beyond* $4 - 2\sqrt{3}$. Do higher-dimensional Goren–Oort configurations, or other algebraic sources of tensor-type collapse, give rates above $4 - 2\sqrt{3}$ on bounded-degree relations with the paired-distance property?

(6) *Characteristic zero.* Do linear codes over fields of characteristic zero obey the half bound on all bounded-degree relations with the paired-distance property, without the coefficient and width bounds of the fifth paper's complex-linear theorem (Remark 10.3)?

A Index of Lean declarations

The status lines of the paper cite the following declarations, listed by Lean module with their fully qualified names. Module files are relative to the paper's `lean/` directory. The axiom checks in `lean/PaperAxioms.lean` are generated from the same sources.

`AboveHalfRate.BlowUp` `AboveHalf.blowCode`, `AboveHalf.blowUp`, `AboveHalf.card_blowCode`, `AboveHalf.degree_blowUp_le`, `AboveHalf.edist_le_blowUp`, `AboveHalf.isStorageCode_blowCode`, `AboveHalf.shortPairedCoordinates_blowUp_subset`.

AboveHalfRate.Descent AboveHalf.CheckSystem.card_in_qrel_le, AboveHalf.CheckSystem.card_qrel_le, AboveHalf.CheckSystem.degree_symGraph_le, AboveHalf.CheckSystem.isStorageCode_nbhd_symGraph, AboveHalf.CheckSystem.isStorageCode_qrel, AboveHalf.CheckSystem.le_mul_of_edist_le, AboveHalf.CheckSystem.le_mul_of_odd_walk, AboveHalf.CheckSystem.proj_not_mem_qrel, AboveHalf.CheckSystem.qrel, AboveHalf.CheckSystem.symGraph.

AboveHalfRate.Fluid AboveHalf.Fluid.charges_sum, AboveHalf.Fluid.f0_le_half, AboveHalf.Fluid.f0_nonneg, AboveHalf.Fluid.f1_le_one, AboveHalf.Fluid.f1_nonneg, AboveHalf.Fluid.integral_f0, AboveHalf.Fluid.integral_f0_sq_le, AboveHalf.Fluid.integral_f0_sq_sub_le, AboveHalf.Fluid.integral_f1, AboveHalf.Fluid.integral_f1_sq_sub_le, AboveHalf.Fluid.one_sub_tau_sq.

AboveHalfRate.LiftSelection AboveHalf.IsDegenerateFor, AboveHalf.IsEarlierEnd, AboveHalf.IsLift, AboveHalf.IsLift.map_liftSwap, AboveHalf.exists_assignment_of_fractional, AboveHalf.exists_liftSelection_of_fluid, AboveHalf.isDegenerateFor_image_liftOf, AboveHalf.isEarlierEnd_liftOf, AboveHalf.isEarlierEnd_lifts, AboveHalf.isLift_iff, AboveHalf.isOption_iff_exists_lift, AboveHalf.isOption_mk_iff, AboveHalf.liftOf_options, AboveHalf.liftPos_injective, AboveHalf.liftSwap.

AboveHalfRate.Main AboveHalf.not_Q91At_of_tower, AboveHalf.not_Q91BinaryAt_of_tower.

AboveHalfRate.Numbers AboveHalf.Numbers.base_bernstein, AboveHalf.Numbers.base_degeneracy.

AboveHalfRate.Peeling AboveHalf.exists_private_check, AboveHalf.exists_walk_of_down, AboveHalf.support_downChecks.

AboveHalfRate.Tanner AboveHalf.interpolates_of_minWeight, AboveHalf.pairing_eq_zero_of_mem_downChecks, AboveHalf.pairing_eq_zero_of_mem_iSup.

AboveHalfRate.Undirected AboveHalf.degree_graphBlowUp_le, AboveHalf.exists_walk_of_graphBlowUp, AboveHalf.graphBlowUp, AboveHalf.graphBlowUpHom, AboveHalf.isStorageCode_graphBlowUp, AboveHalf.nbhd_graphBlowUp, AboveHalf.undirected_tower, AboveHalf.undirected_tower_binary.

Also named in the text From this paper's library: AboveHalf.CheckSystem, AboveHalf.Fluid.f0_sub_le_one, AboveHalf.Fluid.f1_sub_le_one, AboveHalf.Numbers.binary_rate, AboveHalf.Numbers.params.
From the eleventh paper's library: MergeDepth.Q91At, MergeDepth.Q91BinaryAt.
From the fifth paper's library: DirectedRecovery.HasPairedDistanceSeparation, DirectedRecovery.pairedIncidenceGraph.
From the fourth paper's library: MotionDepthCodes.IsStorageCode.

B Building and verifying the formalization

The Lean project is the directory `lean/` next to this paper. It uses Lean and Mathlib, both at version v4.34.1 (`lean-toolchain`, `lakefile.toml`, `lake-manifest.json`), and requires the library of the eleventh paper of the series through a relative path; through it, it uses the libraries of the fourth and fifth papers, the shared library `lean-machine` and the other libraries those depend on. The sibling directories of the repository must therefore stay in place.

Build. With `elan` installed, run in `lean/`:

```
lake exe cache get && lake build
```

The first command downloads the compiled Mathlib; the second compiles the library and the libraries it requires. No heartbeat limit is raised; the only option set is `autoImplicit false`.

Axioms. The file `PaperAxioms.lean`, generated from the LaTeX sources of this paper by `tools/lean_index.py`, runs `#print axioms` on every declaration named in the paper (Appendix A):

```
lake env lean PaperAxioms.lean
```

Every declaration depends at most on `propext`, `Classical.choice` and `Quot.sound`. No Lean file of the library uses `sorry`, `admit` or `native_decide`, or declares an `axiom`.

The statement. The target statements are those of the eleventh paper, `MergeDepth.Q91At` and `MergeDepth.Q91BinaryAt`. They are built from three earlier definitions:

- the paired incidence graph `DirectedRecovery.pairedIncidenceGraph` of the fifth paper;
- the paired-distance condition `DirectedRecovery.HasPairedDistanceSeparation` of the fifth paper;
- storage codes, `MotionDepthCodes.IsStorageCode`, of the fourth paper.

The main formal results, `AboveHalf.not_Q91At_of_tower` and `AboveHalf.not_Q91BinaryAt_of_tower`, prove their negations. The first needs a family of separated checks (`AboveHalf.CheckSystem`) and a tower of quotients carrying codes of rate $\rho > 1/2$ annihilated by the descended checks; the second needs in addition an injective encoding of the field in b bits with $\rho \log_2 |\mathbb{F}|/b > 1/2$. They are stated for an abstract set of coordinates with a group action and a distance, rather than for the Bruhat–Tits tree, which is not available in Mathlib; Sections 4 and 5 explain how the tree provides these data.

Where things are. `Fluid.lean` (Lemma 3.5), `LiftSelection.lean` (Definition 3.1, Lemmas 3.3 and 3.4 and the deterministic part of Theorem 3.2), `Peeling.lean` (Lemma 4.1), `Tanner.lean` (Lemmas 2.3 and 2.4), `Descent.lean` (Proposition 5.1 and the symmetrized graphs of Section 9), `Incidence.lean` (degrees and distances in paired incidence graphs), `BlowUp.lean` (Lemma 8.1), `Main.lean` (the formal parts of Theorem 5.3, Theorem 7.1 and Corollary 8.2), `Undirected.lean` (Lemma 9.2 and the formal part of Corollary 9.3) and `Numbers.lean` (the numerical facts of Corollaries 3.7 and 8.2 and Theorem 7.1). `Main.lean` and `Undirected.lean` do not import `Peeling.lean`, `Tanner.lean`, `LiftSelection.lean`, `Fluid.lean` or `Numbers.lean`: the main theorems take the family of checks, the annihilation of the codes and the rate as hypotheses, and the composition of the formal parts is on paper.

References

[1] Ross Atkins, Puck Rombach, and Fiona Skerman. Guessing numbers of odd cycles. *The Electronic Journal of Combinatorics*, 24(1):P1.45, 2017. doi:10.37236/5964.

[2] Eitan Bachmat and Eyal Z. Goren. On the non-ordinary locus in Hilbert–Blumenthal surfaces. *Mathematische Annalen*, 313(3):475–506, 1999. Proposition 4.1 and Theorem 6.1 cited.

[3] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013. Section 2.8 (Bernstein's inequality) cited. doi:10.1093/acprof:oso/9780199535255.001.0001.

[4] Hans Brende and Claude Opus 5.5 (Anthropic). Bounded nesting and depth reduction for matrix transposition on tapes. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision a2c99b545bba08d317b2d3c6df786b6bd6d04217. URL: https://github.com/HansBrende/papers/tree/a2c99b545bba08d317b2d3c6df786b6bd6d04217/Bounded-Nesting-and-Depth-Reduction-for-Matrix-Transposition-on-Tapes.

[5] Hans Brende and Claude Opus 5.5 (Anthropic). Curved motion, depth reduction, and storage codes for matrix transposition on tapes. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision a2c99b545bba08d317b2d3c6df786b6bd6d04217. URL: https://github.com/HansBrende/papers/tree/a2c99b545bba08d317b2d3c6df786b6bd6d04217/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes.

[6] Hans Brende and Claude Opus 5.5 (Anthropic). Lück approximation fails in positive characteristic. Manuscript and Lean formalization, repository HansBrende/papers, November 2026. Pinned revision a2c99b545bba08d317b2d3c6df786b6bd6d04217. URL: https://github.com/HansBrende/papers/tree/a2c99b545bba08d317b2d3c6df786b6bd6d04217/Luck-Approximation-Fails-in-Positive-Characteristic.

[7] Hans Brende and Claude Opus 5.5 (Anthropic). Merge depth, input steering, and the coding core of matrix transposition on tapes. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision a2c99b545bba08d317b2d3c6df786b6bd6d04217. URL: https://github.com/HansBrende/papers/tree/a2c99b545bba08d317b2d3c6df786b6bd6d04217/Merge-Depth-Input-Steering-and-the-Coding-Core-of-Matrix-Transposition-on-Tapes.

[8] Hans Brende and Claude Opus 5.5 (Anthropic). Nested elimination and half-rate bounds for storage codes. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision a2c99b545bba08d317b2d3c6df786b6bd6d04217. URL: https://github.com/HansBrende/papers/tree/a2c99b545bba08d317b2d3c6df786b6bd6d04217/Nested-Elimination-and-Half-Rate-Bounds-for-Storage-Codes.

[9] Hans Brende and Gemini 4 Argon (Google DeepMind). Directed reciprocity and deep-nesting bounds for matrix transposition on tapes. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision a2c99b545bba08d317b2d3c6df786b6bd6d04217. URL: https://github.com/HansBrende/papers/tree/a2c99b545bba08d317b2d3c6df786b6bd6d04217/Directed-Reciprocity-and-Deep-Nesting-Bounds-for-Matrix-Transposition-on-Tapes.

[10] Hans Brende and GPT-6 Astra (OpenAI). Covers and local generation in linear storage codes. Manuscript and partial Lean formalization, repository HansBrende/papers, October 2026. Pinned revision a2c99b545bba08d317b2d3c6df786b6bd6d04217. URL: https://github.com/HansBrende/papers/tree/a2c99b545bba08d317b2d3c6df786b6bd6d04217/Covers-and-Local-Generation-in-Linear-Storage-Codes.

[11] Hans Brende and GPT-6 Astra (OpenAI). Directed recovery structures in tape computations. Manuscript and partial Lean formalization, repository HansBrende/papers, October 2026. Pinned revision a2c99b545bba08d317b2d3c6df786b6bd6d04217. URL: https://github.com/HansBrende/papers/tree/a2c99b545bba08d317b2d3c6df786b6bd6d04217/Directed-Recovery-Structures-in-Tape-Computations.

[12] Ching-Li Chai. Arithmetic minimal compactification of the Hilbert–Blumenthal moduli spaces. *Annals of Mathematics (2)*, 131(3):541–554, 1990.

[13] Demetres Christofides and Klas Markström. The guessing number of undirected graphs. *The Electronic Journal of Combinatorics*, 18(1):P192, 2011. doi:10.37236/679.

[14] Bruno Codenotti, Pavel Pudlák, and Giovanni Resta. Some structural properties of low-rank matrices related to computational complexity. *Theoretical Computer Science*, 235(1):89–107, 2000. doi:10.1016/S0304-3975(99)00185-1.

[15] Mladen Dimitrov. Compactifications arithmétiques des variétés de Hilbert et formes modulaires de Hilbert pour $\Gamma_1(\mathfrak{c}, \mathfrak{n})$. In *Geometric Aspects of Dwork Theory*, pages 527–554. Walter de Gruyter, Berlin, 2004. Section 7 and Theorem 8.3 cited (numbering of arXiv:math/0212071v3).

[16] Alexander Golovnev and Ishay Haviv. The (generalized) orthogonality dimension of (generalized) Kneser graphs: Bounds and applications. *Theory of Computing*, 18(22):1–22, 2022. Preprint arXiv:2002.08580; Theorem 1.10 cited (same number in arXiv versions 1 and 2 and in the journal). doi:10.4086/toc.2022.v018a022.

[17] Eyal Z. Goren. Hasse invariants for Hilbert modular varieties. *Israel Journal of Mathematics*, 122:157–174, 2001. Section 1 and Theorem 2.1 cited.

[18] P. Hall. On representatives of subsets. *Journal of the London Mathematical Society*, s1-10(1):26–30, 1935. doi:10.1112/jlms/s1-10.37.26.

[19] Ishay Haviv. On minrank and forbidden subgraphs. *ACM Transactions on Computation Theory*, 11(4), 2019. Preprint arXiv:1806.00638. doi:10.1145/3322817.

[20] Friedrich Hirzebruch. Hilbert modular surfaces. *L'Enseignement Mathématique (2)*, 19:183–281, 1973. Sections 1.6, 1.7, 2.5, 2.6, 3.2 and 3.6 cited.

[21] Andrei Jaikin-Zapirain. L^2-Betti numbers and their analogues in positive characteristic. In *Groups St Andrews 2017 in Birmingham*, volume 455 of *London Mathematical Society Lecture Note Series*, pages 346–405. Cambridge University Press, 2019. Section 1 and Theorem 10.1 cited. doi:10.1017/9781108692397.015.

[22] Andrei Jaikin-Zapirain and Henrique Souza. Sylvester domains and pro-p groups. *Documenta Mathematica*, 31(2):453–502, 2026. Preprint arXiv:2402.14130; Theorem 8.1 cited. doi:10.4171/DM/1034.

[23] Swastik Kopparty and Itzhak Tamo. Algebraic expander codes. Preprint arXiv:2603.24788, 2026. URL: https://arxiv.org/abs/2603.24788.

[24] Stephen S. Kudla and Michael Rapoport. Arithmetic Hirzebruch–Zagier cycles. *Journal für die reine und angewandte Mathematik*, 515:155–244, 1999. Section 4, (4.3), (4.5), (4.6),

Definition 4.1, Lemma 4.3, (4.9) and the text after it, Proposition 4.4 and (5.12) cited (numbering of arXiv:math/9904083).

[25] Kai-Wen Lan. *Arithmetic Compactifications of PEL-Type Shimura Varieties*, volume 36 of *London Mathematical Society Monographs Series*. Princeton University Press, 2013. Theorems 6.4.1.1 and 7.3.3.4 and Definitions 7.3.1.1 and 7.3.1.3 cited (numbering checked in Lan's 2008 Harvard thesis, on which the book is based).

[26] Christian Liedtke. Algebraic surfaces in positive characteristic. In Fedor Bogomolov, Brendan Hassett, and Yuri Tschinkel, editors, *Birational Geometry, Rational Curves, and Arithmetic*, pages 229–292. Springer, New York, 2013. Preprint arXiv:0912.4291; Section 4.1 cited. doi:10.1007/978-1-4614-6482-2_11.

[27] A. Lubotzky, R. Phillips, and P. Sarnak. Ramanujan graphs. *Combinatorica*, 8(3):261–277, 1988. doi:10.1007/BF02126799.

[28] Wolfgang Lück. Approximating L^2-invariants by their finite-dimensional analogues. *Geometric and Functional Analysis*, 4(4):455–481, 1994.

[29] Arya Mazumdar. Storage capacity of repairable networks. *IEEE Transactions on Information Theory*, 61(11):5810–5821, 2015. doi:10.1109/TIT.2015.2472521.

[30] Arya Mazumdar, Andrew McGregor, and Sofya Vorotnikova. Storage capacity as an information-theoretic vertex cover and the index coding rate. *IEEE Transactions on Information Theory*, 65(9):5580–5591, 2019. Preprint arXiv:1706.09197; Theorem 9 cited (numbering of arXiv versions 2 and 3). doi:10.1109/TIT.2019.2910026.

[31] Vladimir Platonov and Andrei Rapinchuk. *Algebraic Groups and Number Theory*, volume 139 of *Pure and Applied Mathematics*. Academic Press, 1994. Chapter 6 and Section 7.4 cited.

[32] Michael Rapoport. Compactifications de l'espace de modules de Hilbert–Blumenthal. *Compositio Mathematica*, 36(3):255–335, 1978. Theorem 5.1, Proposition 5.2 and Corollary 5.3 cited.

[33] Michael Rapoport and Thomas Zink. *Period Spaces for p-Divisible Groups*, volume 141 of *Annals of Mathematics Studies*. Princeton University Press, 1996. Theorem 6.30 cited.

[34] Jean-Pierre Serre. *Trees*. Springer, 1980. Translated from the French by John Stillwell; Section I.4 cited.

[35] Hellmuth Stamm. On the reduction of the Hilbert–Blumenthal-moduli scheme with $\Gamma_0(p)$-level structure. *Forum Mathematicum*, 9(4):405–455, 1997.

[36] R. Michael Tanner. A recursive approach to low complexity codes. *IEEE Transactions on Information Theory*, 27(5):533–547, 1981. doi:10.1109/TIT.1981.1056404.

[37] John Tate. Endomorphisms of abelian varieties over finite fields. *Inventiones Mathematicae*, 2:134–144, 1966.

[38] Yichao Tian and Liang Xiao. On Goren–Oort stratification for quaternionic Shimura varieties. *Compositio Mathematica*, 152(10):2134–2220, 2016.

[39] M. A. Tsfasman, S. G. Vlăduţ, and Th. Zink. Modular curves, Shimura curves, and Goppa codes, better than Varshamov–Gilbert bound. *Mathematische Nachrichten*, 109(1):21–28, 1982. doi:10.1002/mana.19821090103.

[40] Marie-France Vignéras. *Arithmétique des algèbres de quaternions*, volume 800 of *Lecture Notes in Mathematics*. Springer, 1980. Corollaire I.4.11, Theorem III.4.1, Proposition V.3.4 and the table on p. 155 cited.